model_2500 · All 11 actual policy observations · env 8 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 8
DR · 0.00–60.00 s

MuJoCo · physics/PD 400 Hz · env 8
DR · 0.00–60.00 s

MuJoCo · physics/PD 800 Hz · env 8
DR · 0.00–60.00 s

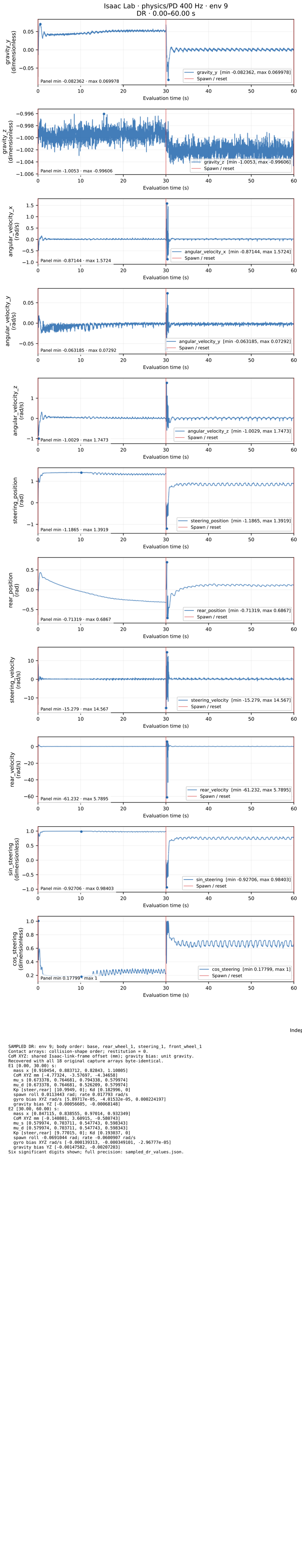
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

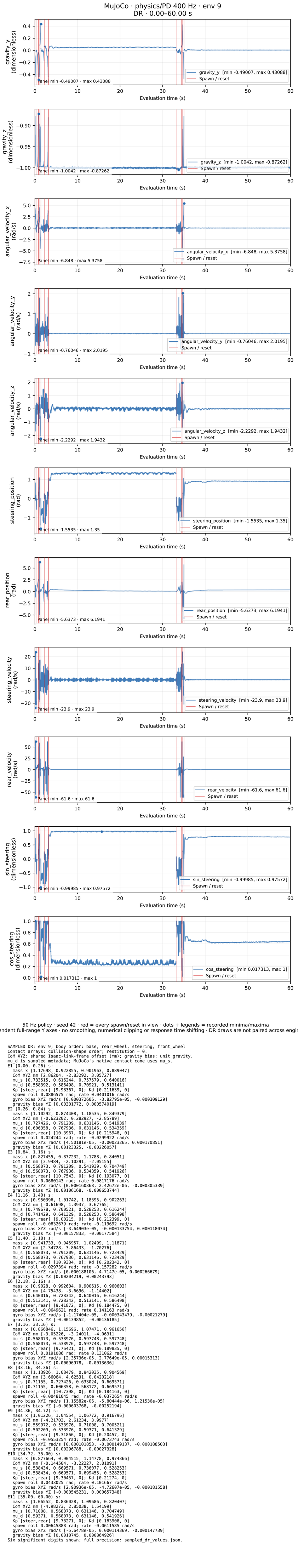
```
SAMPLED DR: env 8; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.05978, 0.906995, 0.869317, 1.14259]
  CoM XYZ mm [1.60898, 1.46585, -1.6959]
  mu_s [0.673378, 0.531594, 0.741039, 0.713129]
  mu_d [0.673378, 0.531594, 0.614405, 0.713129]
  Kp [steer,rear] [9.69437, 0]; Kd [0.203353, 0]
  spawn roll 0.0265545 rad; rate -0.000518277 rad/s
  gyro bias XYZ rad/s [0.000203254, -0.000297841, 0.000206812]
  gravity bias YZ [-0.00221992, 0.000585724]
E2 [30.00, 60.00) s:
  mass x [1.10196, 0.803254, 0.871098, 1.15485]
  CoM XYZ mm [2.80611, 3.00247, 0.461772]
  mu_s [0.533455, 0.751971, 0.741039, 0.576972]
  mu_d [0.533455, 0.541224, 0.614405, 0.576972]
  Kp [steer,rear] [9.27546, 0]; Kd [0.203981, 0]
  spawn roll 0.0704735 rad; rate -0.18156 rad/s
  gyro bias XYZ rad/s [-0.000119344, 0.0002309, 5.98212e-06]
  gravity bias YZ [-0.00216605, -0.00250907]
Six significant digits shown; full precision: sampled_dr_values.json.
```

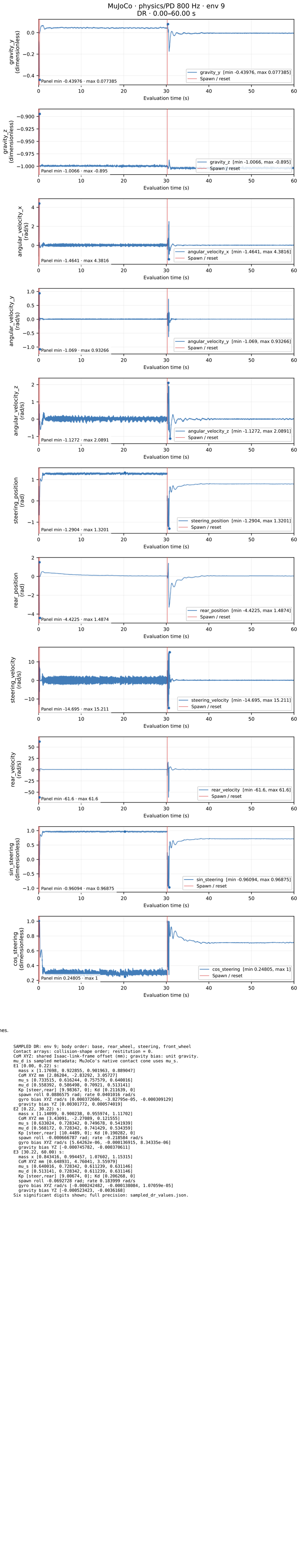
```
SAMPLED DR: env 8; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.46) s:
  mass x [1.11497, 1.13347, 1.01916, 1.18938]
  CoM XYZ mm [-2.63166, 1.46923, -4.34974]
  mu_s [0.700521, 0.640867, 0.732187, 0.733515]
  mu_d [0.641329, 0.556841, 0.631664, 0.558392]
  Kp [steer,rear] [9.53193, 0]; Kd [0.195893, 0]
  spawn roll -0.096984 rad; rate 0.0498493 rad/s
  gyro bias XYZ rad/s [3.95927e-07, -0.000539962, -0.000311445]
  gravity bias YZ [-0.000585585, 0.000518127]
E2 [0.46, 0.88) s:
  mass x [0.971118, 0.848327, 1.18653, 1.02509]
  CoM XYZ mm [-0.876627, 4.23962, 3.52719]
  mu_s [0.733515, 0.704749, 0.546287, 0.528253]
  mu_d [0.558392, 0.541926, 0.546287, 0.528253]
  Kp [steer,rear] [9.90465, 0]; Kd [0.198417, 0]
  spawn roll -0.0590784 rad; rate -0.224446 rad/s
  gyro bias XYZ rad/s [-5.93493e-05, 0.000112362, 0.000246449]
  gravity bias YZ [0.000639852, -0.000516005]
E3 [0.88, 30.88) s:
  mass x [0.860593, 0.937091, 1.18449, 0.958447]
  CoM XYZ mm [-0.61012, 0.384081, -4.92167]
  mu_s [0.611239, 0.71155, 0.757579, 0.538976]
  mu_d [0.611239, 0.71155, 0.70921, 0.538976]
  Kp [steer,rear] [10.827, 0]; Kd [0.191736, 0]
  spawn roll 0.0245912 rad; rate -0.132646 rad/s
  gyro bias XYZ rad/s [0.000117953, -9.70442e-06, 0.00021612]
  gravity bias YZ [-0.00107207, -0.00201265]
E4 [30.88, 31.22) s:
  mass x [1.10668, 1.04798, 0.924135, 1.16744]
  CoM XYZ mm [1.7946, -4.40723, -3.96957]
  mu_s [0.538976, 0.640867, 0.637675, 0.559972]
  mu_d [0.538976, 0.556841, 0.637675, 0.502209]
  Kp [steer,rear] [10.423, 0]; Kd [0.188921, 0]
  spawn roll 0.0447315 rad; rate -0.139914 rad/s
  gyro bias XYZ rad/s [0.000180307, -0.000155625, 6.8454e-05]
  gravity bias YZ [0.00154868, 0.000783916]
E5 [31.22, 60.00) s:
  mass x [0.939216, 0.816572, 0.856842, 1.18614]
  CoM XYZ mm [1.68749, -0.36036, 2.17776]
  mu_s [0.559972, 0.71155, 0.640867, 0.71155]
  mu_d [0.502209, 0.71155, 0.556841, 0.71155]
  Kp [steer,rear] [9.74387, 0]; Kd [0.19657, 0]
  spawn roll -0.0473665 rad; rate 0.161744 rad/s
  gyro bias XYZ rad/s [9.02701e-05, -0.000169255, 0.000344843]
  gravity bias YZ [0.00157456, 0.000492216]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 8; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.42) s:
  mass x [1.11497, 1.13347, 1.01916, 1.18938]
  CoM XYZ mm [-2.63166, 1.46923, -4.34974]
  mu_s [0.700521, 0.640867, 0.732187, 0.733515]
  mu_d [0.641329, 0.556841, 0.631664, 0.558392]
  Kp [steer,rear] [9.53193, 0]; Kd [0.195893, 0]
  spawn roll -0.096984 rad; rate 0.0498493 rad/s
  gyro bias XYZ rad/s [3.95927e-07, -0.000539962, -0.000311445]
  gravity bias YZ [-0.000585585, 0.000518127]
E2 [0.42, 30.42) s:
  mass x [0.838724, 0.932716, 1.01168, 0.832511]
  CoM XYZ mm [-0.979841, -2.97601, -0.845064]
  mu_s [0.597748, 0.71008, 0.559972, 0.541939]
  mu_d [0.597748, 0.59371, 0.502209, 0.534359]
  Kp [steer,rear] [9.30986, 0]; Kd [0.208562, 0]
  spawn roll 0.0204349 rad; rate 0.232107 rad/s
  gyro bias XYZ rad/s [-0.000214265, -8.75294e-05, -0.00010339]
  gravity bias YZ [0.00236122, -0.000522518]
E3 [30.42, 60.00) s:
  mass x [1.19767, 1.13834, 0.819637, 1.1911]
  CoM XYZ mm [0.542863, 3.11743, -1.37944]
  mu_s [0.71008, 0.559972, 0.640016, 0.541939]
  mu_d [0.59371, 0.502209, 0.513141, 0.534359]
  Kp [steer,rear] [9.8392, 0]; Kd [0.199225, 0]
  spawn roll -0.00352456 rad; rate -0.203906 rad/s
  gyro bias XYZ rad/s [0.000280388, -0.000163418, -0.00011797]
  gravity bias YZ [0.00312605, 0.000531834]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 9 · full 60 s







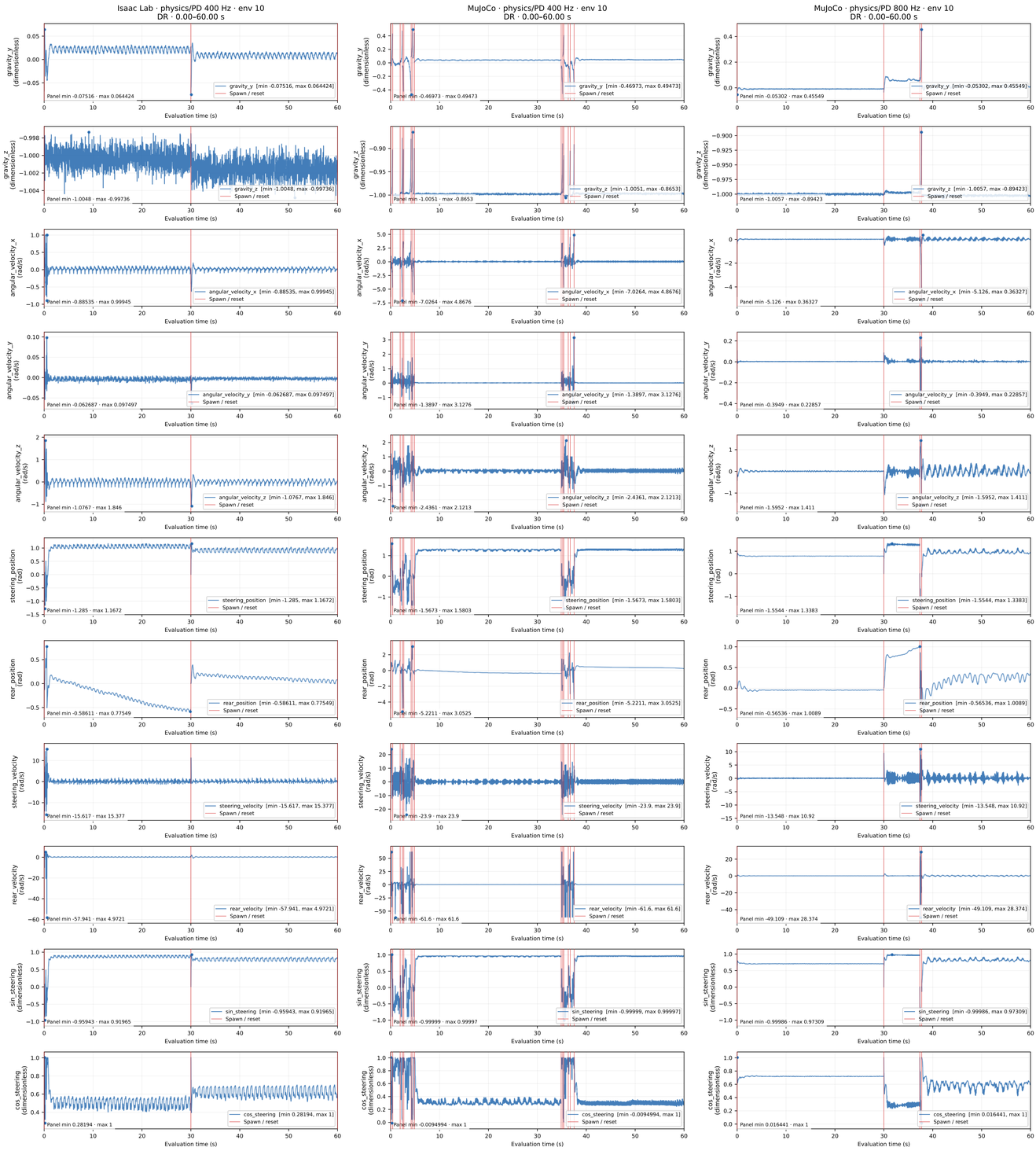
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 9; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.910454, 0.883712, 0.82843, 1.10805]
  CoM XYZ mm [-4.77324, -3.57697, -4.34658]
  mu_s [0.673378, 0.764681, 0.794338, 0.579974]
  mu_d [0.673378, 0.764681, 0.526209, 0.579974]
  Kp [steer,rear] [10.9949, 0]; Kd [0.182996, 0]
  spawn roll 0.0113443 rad; rate 0.017793 rad/s
  gyro bias XYZ rad/s [5.89717e-05, -4.01532e-05, 0.000224197]
  gravity bias YZ [-0.00056605, -0.00068148]
E2 [30.00, 60.00) s:
  mass x [0.847115, 0.838555, 0.97014, 0.932349]
  CoM XYZ mm [-0.140801, 3.60915, -0.580743]
  mu_s [0.579974, 0.703711, 0.547743, 0.598343]
  mu_d [0.579974, 0.703711, 0.547743, 0.598343]
  Kp [steer,rear] [9.77015, 0]; Kd [0.193037, 0]
  spawn roll -0.0691044 rad; rate -0.0600907 rad/s
  gyro bias XYZ rad/s [-0.000139313, -0.000349101, -2.96777e-05]
  gravity bias YZ [-0.00147582, -0.00207203]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 9; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.26) s:
  mass x [1.17698, 0.922855, 0.901963, 0.889047]
  CoM XYZ mm [2.86204, -2.83292, 3.05727]
  mu_s [0.733515, 0.616244, 0.757579, 0.640016]
  mu_d [0.558392, 0.586498, 0.70921, 0.513141]
  Kp [steer,rear] [9.98367, 0]; Kd [0.211639, 0]
  spawn roll 0.0886575 rad; rate 0.0401016 rad/s
  gyro bias XYZ rad/s [0.000372606, -3.82795e-05, -0.000309129]
  gravity bias YZ [0.00301772, 0.000574019]
E2 [0.26, 0.84) s:
  mass x [1.10292, 0.874408, 1.18535, 0.849379]
  CoM XYZ mm [-0.623202, 0.282927, -2.85789]
  mu_s [0.727426, 0.791209, 0.631146, 0.541939]
  mu_d [0.606358, 0.767936, 0.631146, 0.534359]
  Kp [steer,rear] [10.3967, 0]; Kd [0.215948, 0]
  spawn roll 0.024244 rad; rate -0.0299922 rad/s
  gyro bias XYZ rad/s [4.50181e-05, -0.00023265, 0.000170851]
  gravity bias YZ [0.00123325, -0.00226057]
E3 [0.84, 1.16) s:
  mass x [0.827455, 0.877232, 1.1788, 0.84051]
  CoM XYZ mm [3.9484, -2.10291, -2.05155]
  mu_s [0.568073, 0.791209, 0.541939, 0.704749]
  mu_d [0.568073, 0.767936, 0.534359, 0.541926]
  Kp [steer,rear] [10.7543, 0]; Kd [0.193077, 0]
  spawn roll 0.0680143 rad; rate 0.0817176 rad/s
  gyro bias XYZ rad/s [0.000160368, 2.42672e-06, -0.000305339]
  gravity bias YZ [0.00106168, -0.000653744]
E4 [1.16, 1.40) s:
  mass x [0.950396, 1.01742, 1.18395, 0.902263]
  CoM XYZ mm [-0.61698, 1.3937, 3.67765]
  mu_s [0.749678, 0.700521, 0.528253, 0.616244]
  mu_d [0.741429, 0.641329, 0.528253, 0.586498]
  Kp [steer,rear] [9.00215, 0]; Kd [0.212399, 0]
  spawn roll -0.0832679 rad; rate -0.119692 rad/s
  gyro bias XYZ rad/s [-3.64903e-05, -0.000133754, 0.000118074]
  gravity bias YZ [-0.00157833, -0.00177584]
E5 [1.40, 2.18) s:
  mass x [0.941733, 0.945957, 1.02499, 1.11871]
  CoM XYZ mm [2.34728, 3.86433, -1.70276]
  mu_s [0.568073, 0.791209, 0.631146, 0.723429]
  mu_d [0.568073, 0.767936, 0.631146, 0.723429]
  Kp [steer,rear] [10.9334, 0]; Kd [0.202342, 0]
  spawn roll -0.0297394 rad; rate -0.157282 rad/s
  gyro bias XYZ rad/s [0.000188106, 4.7147e-05, 0.000266679]
  gravity bias YZ [0.00204219, 0.00243793]
E6 [2.18, 3.16) s:
  mass x [0.9028, 0.992604, 0.900615, 0.960603]
  CoM XYZ mm [4.75438, -3.6696, -1.14402]
  mu_s [0.640016, 0.728342, 0.640016, 0.616244]
  mu_d [0.513141, 0.728342, 0.513141, 0.586498]
  Kp [steer,rear] [9.41872, 0]; Kd [0.184475, 0]
  spawn roll -0.0649621 rad; rate 0.141163 rad/s
  gyro bias XYZ rad/s [-1.17404e-05, -0.000343479, -0.00021279]
  gravity bias YZ [-0.00139852, -0.00136185]
E7 [3.16, 33.16) s:
  mass x [0.866846, 1.15696, 1.07471, 0.961656]
  CoM XYZ mm [-3.05226, -3.24011, -4.0631]
  mu_s [0.568073, 0.538976, 0.597748, 0.597748]
  mu_d [0.568073, 0.538976, 0.597748, 0.597748]
  Kp [steer,rear] [9.76421, 0]; Kd [0.189835, 0]
  spawn roll 0.0191086 rad; rate 0.131062 rad/s
  gyro bias XYZ rad/s [2.35736e-05, 2.77649e-05, 0.00015313]
  gravity bias YZ [0.00096978, -0.0013636]
E8 [33.16, 34.36) s:
  mass x [1.13926, 1.08479, 0.942035, 0.904569]
  CoM XYZ mm [3.66064, 4.62531, 0.0420218]
  mu_s [0.71155, 0.727426, 0.633024, 0.669571]
  mu_d [0.71155, 0.606358, 0.568172, 0.669571]
  Kp [steer,rear] [10.7398, 0]; Kd [0.184163, 0]
  spawn roll -0.00481045 rad; rate -0.0372654 rad/s
  gyro bias XYZ rad/s [1.15582e-06, -5.80444e-06, 1.21536e-05]
  gravity bias YZ [-0.000603768, -0.00252194]
E9 [34.36, 34.72) s:
  mass x [1.01226, 1.04554, 1.06772, 0.916796]
  CoM XYZ mm [-4.21703, 2.61234, 3.9977]
  mu_s [0.559972, 0.538976, 0.71008, 0.700521]
  mu_d [0.502209, 0.538976, 0.59371, 0.641329]
  Kp [steer,rear] [9.31866, 0]; Kd [0.20457, 0]
  spawn roll -0.0553254 rad; rate -0.0673743 rad/s
  gyro bias XYZ rad/s [0.000101853, -0.000149137, -0.000188503]
  gravity bias YZ [0.00296788, -0.00027328]
E10 [34.72, 35.00) s:
  mass x [0.877664, 0.904515, 1.14778, 0.974366]
  CoM XYZ mm [-0.144504, -3.22227, 2.01091]
  mu_s [0.538434, 0.669571, 0.736077, 0.528253]
  mu_d [0.538434, 0.669571, 0.699455, 0.528253]
  Kp [steer,rear] [9.30457, 0]; Kd [0.21274, 0]
  spawn roll 0.0433025 rad; rate 0.101667 rad/s
  gyro bias XYZ rad/s [2.90936e-05, -4.72607e-06, -0.000181558]
  gravity bias YZ [-0.000545231, 0.000657348]
E11 [35.00, 60.00) s:
  mass x [1.06552, 0.836028, 1.09686, 0.820407]
  CoM XYZ mm [-4.98273, 2.85038, 1.54199]
  mu_s [0.71008, 0.568073, 0.631146, 0.704749]
  mu_d [0.59371, 0.568073, 0.631146, 0.541926]
  Kp [steer,rear] [9.78271, 0]; Kd [0.183908, 0]
  spawn roll 0.00645888 rad; rate -0.0611585 rad/s
  gyro bias XYZ rad/s [-5.6478e-05, 0.000114369, -0.000147739]
  gravity bias YZ [0.0010745, 0.000864926]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 9; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [1.17698, 0.922855, 0.901963, 0.889047]
  CoM XYZ mm [2.86204, -2.83292, 3.05727]
  mu_s [0.733515, 0.616244, 0.757579, 0.640016]
  mu_d [0.558392, 0.586498, 0.70921, 0.513141]
  Kp [steer,rear] [9.98367, 0]; Kd [0.211639, 0]
  spawn roll 0.0886575 rad; rate 0.0401016 rad/s
  gyro bias XYZ rad/s [0.000372606, -3.82795e-05, -0.000309129]
  gravity bias YZ [0.00301772, 0.000574019]
E2 [0.22, 30.22) s:
  mass x [1.14099, 0.900238, 0.955974, 1.11702]
  CoM XYZ mm [3.43091, -2.27089, 0.121555]
  mu_s [0.633024, 0.728342, 0.749678, 0.541939]
  mu_d [0.568172, 0.728342, 0.741429, 0.534359]
  Kp [steer,rear] [10.4489, 0]; Kd [0.190282, 0]
  spawn roll -0.000666787 rad; rate -0.218584 rad/s
  gyro bias XYZ rad/s [5.64262e-06, -0.000136915, 8.34335e-06]
  gravity bias YZ [-0.000745782, -0.000370611]
E3 [30.22, 60.00) s:
  mass x [0.843416, 0.994457, 1.07602, 1.15315]
  CoM XYZ mm [0.648931, 4.76041, 3.55979]
  mu_s [0.640016, 0.728342, 0.611239, 0.631146]
  mu_d [0.513141, 0.728342, 0.611239, 0.631146]
  Kp [steer,rear] [9.00674, 0]; Kd [0.206268, 0]
  spawn roll -0.0692728 rad; rate 0.183999 rad/s
  gyro bias XYZ rad/s [-0.000242482, -0.000138084, 1.07059e-05]
  gravity bias YZ [-0.000523423, -0.0036168]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 10 · full 60 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 10; body order: base, rear_wheel_l, steering_l, front_wheel_l
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.044, 0.937751, 1.19408, 0.987669]
  CoM XYZ mm [-0.877663, 0.708646, 3.21598]
  mu_d [0.531594, 0.560535, 0.764681, 0.674963]
  mu_d [0.531594, 0.560535, 0.764681, 0.601294]
  Kp [steer,rear] [10.1063, 0]; Kd [0.216217, 0]
  spawn roll -0.0612238 rad; rate 0.0984062 rad/s
  gyro bias XYZ rad/s [0.00027079, 0.000114682, 0.000188098]
  gravity bias YZ [0.000185434, -0.000662592]
E2 [30.00, 60.00) s:
  mass x [0.897797, 0.967002, 0.894196, 1.05703]
  CoM XYZ mm [0.710489, 2.68479, -3.00689]
  mu_s [0.602694, 0.579974, 0.760821, 0.741839]
  mu_d [0.602694, 0.579974, 0.670315, 0.614405]
  Kp [steer,rear] [10.8715, 0]; Kd [0.202183, 0]
  spawn roll 0.0764369 rad; rate 0.0505117 rad/s
  gyro bias XYZ rad/s [0.000213677, 0.000031753, 0.000137284]
  gravity bias YZ [0.000929145, -0.00170795]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 10; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.42) s:
  mass x [1.04631, 1.02969, 1.00782, 0.802556]
  CoM XYZ mm [0.502924, 4.68338, -1.0194]
  mu_s [0.728342, 0.732187, 0.666375, 0.633024]
  mu_d [0.728342, 0.633664, 0.519145, 0.568172]
  Kp [steer,rear] [10.8929, 0]; Kd [0.198951, 0]
  spawn roll 0.0530206 rad; rate -0.213743 rad/s
  gyro bias XYZ rad/s [-0.000326526, 0.000265956, -0.000136196]
  gravity bias YZ [-0.00207519, -0.000464771]
E2 [0.42, 1.94) s:
  mass x [1.18486, 1.18667, 0.997063, 1.10972]
  CoM XYZ mm [1.44273, 1.65362, -4.23642]
  mu_s [0.633146, 0.69316, 0.546287, 0.69316]
  mu_d [0.633146, 0.69316, 0.546287, 0.693161]
  Kp [steer,rear] [10.2304, 0]; Kd [0.197478, 0]
  spawn roll -0.0587185 rad; rate 0.0133716 rad/s
  gyro bias XYZ rad/s [0.000119483, 4.30869e-05, -8.05376e-05]
  gravity bias YZ [-0.00049118, -0.00339959]
E3 [1.94, 2.42) s:
  mass x [0.855414, 0.85745, 0.817507, 0.841165]
  CoM XYZ mm [-2.8191, 2.38297, -2.78244]
  mu_s [0.6400916, 0.546287, 0.538434, 0.7234329]
  mu_d [0.513141, 0.546287, 0.538434, 0.7234329]
  Kp [steer,rear] [10.512, 0]; Kd [0.188721, 0]
  spawn roll 0.00295433 rad; rate 0.1409944 rad/s
  gyro bias XYZ rad/s [-2.16887e-05, -5.15203e-05, -0.000142702]
  gravity bias YZ [-0.00417387, 0.0026808]
E4 [2.42, 2.68) s:
  mass x [1.05905, 1.03787, 1.00903, 1.07727]
  CoM XYZ mm [3.46498, -2.57296, 1.03129]
  mu_s [0.7005211, 0.7047149, 0.791209, 0.5599721]
  mu_d [0.6413329, 0.5415926, 0.7677936, 0.5502299]
  Kp [steer,rear] [9.35022, 0]; Kd [0.215744, 0]
  spawn roll 0.0582089 rad; rate 0.0912731 rad/s
  gyro bias XYZ rad/s [0.000100303, 5.99222e-05, 0.000117773]
  gravity bias YZ [-0.00461448, 0.000453434]
E5 [2.68, 4.20) s:
  mass x [1.04755, 0.8141898, 0.9571148, 0.84013]
  CoM XYZ mm [0.49032, -1.96357, -4.73473]
  mu_s [0.7912099, 0.71008, 0.616244, 0.5389756]
  mu_d [0.7677936, 0.59371, 0.586498, 0.5389756]
  Kp [steer,rear] [10.3746, 0]; Kd [0.184485, 0]
  spawn roll 0.0022097 rad; rate -0.0359523 rad/s
  gyro bias XYZ rad/s [-3.85463e-05, -0.000203228, -0.000210508]
  gravity bias YZ [0.000717845, 0.00190448]
E6 [4.20, 4.50) s:
  mass x [0.9702885, 1.00658, 1.07616, 1.04332]
  CoM XYZ mm [3.39187, -0.2957833, 2.05101]
  mu_s [0.723429, 0.69316, 0.791209, 0.7274226]
  mu_d [0.723429, 0.69316, 0.767936, 0.6066358]
  Kp [steer,rear] [10.9528, 0]; Kd [0.205301, 0]
  spawn roll -0.0915221 rad; rate 0.0388739 rad/s
  gyro bias XYZ rad/s [-0.000200333, 5.84175e-05, 0.000113706]
  gravity bias YZ [-0.00230579, 0.002776318]
E7 [4.50, 4.88) s:
  mass x [0.9130352, 1.11761, 0.903278, 0.945502]
  CoM XYZ mm [0.9137318, -0.601137, 0.002656321]
  mu_s [0.6162644, 0.631146, 0.666375, 0.7321827]
  mu_d [0.5864908, 0.631146, 0.519145, 0.631664]
  Kp [steer,rear] [9.18786, 0]; Kd [0.20778, 0]
  spawn roll -0.0728771 rad; rate 0.1531996 rad/s
  gyro bias XYZ rad/s [6.28892e-06, -0.000165498, -4.58954e-05]
  gravity bias YZ [-0.00158328, -0.00160853]
E8 [4.88, 34.88) s:
  mass x [1.0694, 1.11081, 1.11693, 0.99577]
  CoM XYZ mm [-0.171862, -2.03855, 2.83541]
  mu_s [0.6112239, 0.6162644, 0.700521, 0.693161]
  mu_d [0.6112239, 0.5864908, 0.641329, 0.693161]
  Kp [steer,rear] [9.02021, 0]; Kd [0.205425, 0]
  spawn roll -0.09931612 rad; rate -0.125238 rad/s
  gyro bias XYZ rad/s [-5.01276e-05, 0.000197057, -0.000459944]
  gravity bias YZ [0.000572227, 0.001037277]
E9 [34.88, 35.18) s:
  mass x [0.863832, 0.992903, 1.07255, 0.921582]
  CoM XYZ mm [0.823676, 0.6139356, -4.037092]
  mu_s [0.6112239, 0.727426, 0.546287, 0.7321827]
  mu_d [0.6112239, 0.606358, 0.546287, 0.631664]
  Kp [steer,rear] [10.5063, 0]; Kd [0.205938, 0]
  spawn roll 0.0703838 rad; rate 0.0852139 rad/s
  gyro bias XYZ rad/s [-5.24537e-05, -0.000353689, -0.000179775]
  gravity bias YZ [-6.7528e-06, 0.00070289]
E10 [35.18, 35.42) s:
  mass x [1.054, 1.14715, 0.863331, 1.16766]
  CoM XYZ mm [0.7946241, 4.213567, -4.365101]
  mu_s [0.69316, 0.7496678, 0.541939, 0.7335151]
  mu_d [0.69316, 0.741429, 0.534599, 0.5583922]
  Kp [steer,rear] [10.509, 0]; Kd [0.186977, 0]
  spawn roll -0.0988226 rad; rate -0.0518785 rad/s
  gyro bias XYZ rad/s [-0.000270208, -0.000125109, 0.000176881]
  gravity bias YZ [0.000225012, 0.000489124]
E11 [35.42, 36.32) s:
  mass x [1.122, 1.00575, 1.14129, 1.14338]
  CoM XYZ mm [2.424045, 4.91197, -4.948331]
  mu_s [0.5680873, 0.7496678, 0.6408057, 0.7912099]
  mu_d [0.5680873, 0.741429, 0.5568041, 0.7677936]
  Kp [steer,rear] [9.3717, 0]; Kd [0.185427, 0]
  spawn roll -0.0492544 rad; rate 0.0477335 rad/s
  gyro bias XYZ rad/s [0.000381953, 4.79395e-05, -0.000328948]
  gravity bias YZ [0.000363512, -0.00270865]
E12 [36.32, 36.42) s:
  mass x [1.00643, 0.9224027, 1.14794, 0.9653641]
  CoM XYZ mm [-1.756857, 2.66316, 0.1633766]
  mu_s [0.7047149, 0.7285325, 0.727426, 0.6112239]
  mu_d [0.541926, 0.728342, 0.606358, 0.6112239]
  Kp [steer,rear] [9.20986, 0]; Kd [0.201713, 0]
  spawn roll 0.0762615 rad; rate 0.0145723 rad/s
  gyro bias XYZ rad/s [-0.000218343, 4.74383e-05, 0.000106282]
  gravity bias YZ [0.00176086, -0.00215085]
E13 [36.42, 37.22) s:
  mass x [0.9279227, 1.08205, 1.0375, 0.9140039]
  CoM XYZ mm [3.29695, 2.21762, 1.84489]
  mu_s [0.704749, 0.538976, 0.669571, 0.6376751]
  mu_d [0.541926, 0.538976, 0.669571, 0.6376751]
  Kp [steer,rear] [10.5304, 0]; Kd [0.191862, 0]
  spawn roll -0.0204232 rad; rate -0.0546812 rad/s
  gyro bias XYZ rad/s [0.000184849, -0.000153562, 0.000158029]
  gravity bias YZ [0.000199321, -0.00144308]
E14 [37.22, 37.52) s:
  mass x [1.127, 1.12193, 0.976452, 0.83567, 1.01326]
  CoM XYZ mm [-2.268701, -2.56581, -1.691593]
  mu_s [0.597748, 0.718008, 0.6440816, 0.6408816]
  mu_d [0.597748, 0.59371, 0.513141, 0.5131413]
  Kp [steer,rear] [9.822995, 0]; Kd [0.185461, 0]
  spawn roll 0.0371648 rad; rate 0.0142472 rad/s
  gyro bias XYZ rad/s [0.000117595, -0.000168873, -0.000143787]
  gravity bias YZ [0.000274915, 0.001078559]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 10; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04631, 1.02969, 1.00782, 0.802556]
  CoM XYZ mm [0.502924, 4.68338, -1.0194]
  mu_s [0.728342, 0.732187, 0.666375, 0.633024]
  mu_d [0.728342, 0.633664, 0.519145, 0.568172]
  Kp [steer,rear] [10.8929, 0]; Kd [0.198951, 0]
  spawn roll 0.0530206 rad; rate -0.213743 rad/s
  gyro bias XYZ rad/s [-0.000326526, 0.000265956, -0.000136196]
  gravity bias YZ [-0.00207519, -0.000464771]
E2 [30.00, 37.38) s:
  mass x [0.8090685, 1.1514, 0.91927, 1.0749]
  CoM XYZ mm [1.54605, -4.9049, -1.17566]
  mu_s [0.6331146, 0.727426, 0.538434, 0.6112239]
  mu_d [0.6331146, 0.606358, 0.538434, 0.6112239]
  Kp [steer,rear] [10.5435, 0]; Kd [0.188406, 0]
  spawn roll 0.03338038 rad; rate 0.1337116 rad/s
  gyro bias XYZ rad/s [-0.000131766, 0.000184947, -1.18163e-05]
  gravity bias YZ [-0.00112885, 0.000529576]
E3 [37.38, 37.74) s:
  mass x [0.8562698, 1.1043, 1.09897, 1.1517]
  CoM XYZ mm [-4.89209, 2.17899, -4.68836]
  mu_s [0.7274229, 0.538434, 0.559972, 0.7274226]
  mu_d [0.7274229, 0.538434, 0.502209, 0.6066358]
  Kp [steer,rear] [10.1784, 0]; Kd [0.186521, 0]
  spawn roll -0.0702924 rad; rate -0.1418229 rad/s
  gyro bias XYZ rad/s [0.000132492, 2.44563e-05, -1.87751e-06]
  gravity bias YZ [0.00521048, -0.0000033314]
E4 [37.74, 60.00) s:
  mass x [1.04979, 1.16416, 1.143, 0.8709955]
  CoM XYZ mm [1.27334, 2.53976, -4.629694]
  mu_s [0.546287, 0.757579, 0.640867, 0.7577579]
  mu_d [0.546287, 0.70921, 0.556841, 0.709211]
  Kp [steer,rear] [10.2702, 0]; Kd [0.21008, 0]
  spawn roll 0.0042111 rad; rate -0.1814804 rad/s
  gyro bias XYZ rad/s [-0.000119997, -0.000341104, -0.000298524]
  gravity bias YZ [0.00194827, -0.00207071]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 11 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 11
DR · 0.00–60.00 s

MuJoCo · physics/PD 400 Hz · env 11
DR · 0.00–60.00 s

MuJoCo · physics/PD 800 Hz · env 11
DR · 0.00–60.00 s

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

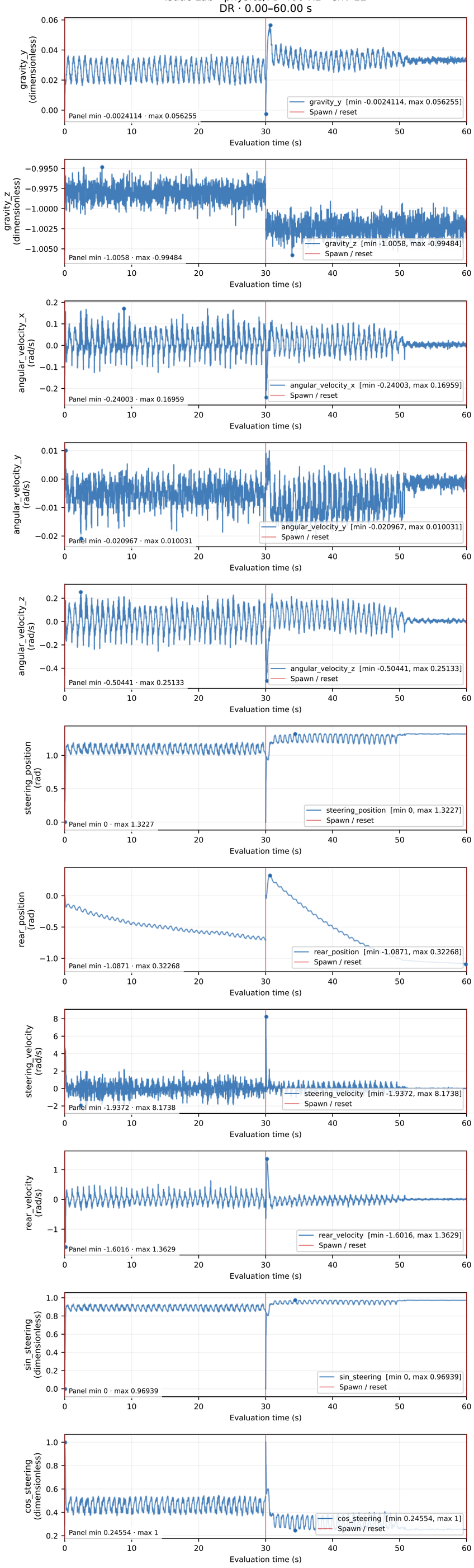
```
SAMPLED DR: env 11; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.16814, 1.08509, 0.842065, 1.12553]
  CoM XYZ mm [1.97017, 0.771385, -4.30979]
  mu_s [0.632409, 0.560535, 0.681711, 0.785466]
  mu_d [0.589076, 0.560535, 0.611756, 0.52258]
  Kp [steer,rear] [9.49828, 0]; Kd [0.189078, 0]
  spawn roll 0.0684242 rad; rate 0.00828874 rad/s
  gyro bias XYZ rad/s [0.000457135, 0.0003747, -0.00031351]
  gravity bias YZ [-0.00394906, 0.00108566]
E2 [30.00, 60.00) s:
  mass x [0.961653, 0.90604, 0.803862, 0.836442]
  CoM XYZ mm [-3.44978, -3.43411, -1.61833]
  mu_s [0.598343, 0.760821, 0.533455, 0.774409]
  mu_d [0.598343, 0.670315, 0.533455, 0.561095]
  Kp [steer,rear] [10.3132, 0]; Kd [0.201799, 0]
  spawn roll 0.0466542 rad; rate -0.224283 rad/s
  gyro bias XYZ rad/s [-6.36372e-05, -0.000145016, 5.87831e-05]
  gravity bias YZ [0.00127278, -0.00158957]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 11; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.40) s:
  mass x [0.804451, 1.08756, 0.932451, 1.17324]
  CoM XYZ mm [-3.95175, 1.66437, 3.19254]
  mu_s [0.69316, 0.749678, 0.616244, 0.669571]
  mu_d [0.69316, 0.741429, 0.586498, 0.669571]
  Kp [steer,rear] [10.5376, 0]; Kd [0.192053, 0]
  spawn roll -0.0839723 rad; rate 0.152834 rad/s
  gyro bias XYZ rad/s [1.32216e-05, -0.000139725, -3.55158e-05]
  gravity bias YZ [0.000531016, -0.00231086]
E2 [0.40, 30.40) s:
  mass x [1.18685, 1.01724, 1.15557, 1.14079]
  CoM XYZ mm [1.51405, 3.86921, -1.64199]
  mu_s [0.546287, 0.631146, 0.611239, 0.727426]
  mu_d [0.546287, 0.631146, 0.611239, 0.606358]
  Kp [steer,rear] [10.0285, 0]; Kd [0.186463, 0]
  spawn roll 0.0861315 rad; rate -0.157324 rad/s
  gyro bias XYZ rad/s [1.12137e-05, -0.000358171, 7.92227e-05]
  gravity bias YZ [-0.000299793, -0.00456061]
E3 [30.40, 60.00) s:
  mass x [0.939248, 0.918164, 1.03497, 0.992503]
  CoM XYZ mm [-3.41698, 3.79475, -2.0653]
  mu_s [0.71155, 0.727426, 0.727426, 0.727426]
  mu_d [0.71155, 0.606358, 0.606358, 0.606358]
  Kp [steer,rear] [9.41325, 0]; Kd [0.201469, 0]
  spawn roll -0.0253665 rad; rate -0.0987004 rad/s
  gyro bias XYZ rad/s [-0.000147457, -7.24679e-05, 0.000179976]
  gravity bias YZ [-0.00232873, -0.000743975]
Six significant digits shown; full precision: sampled_dr_values.json.
```

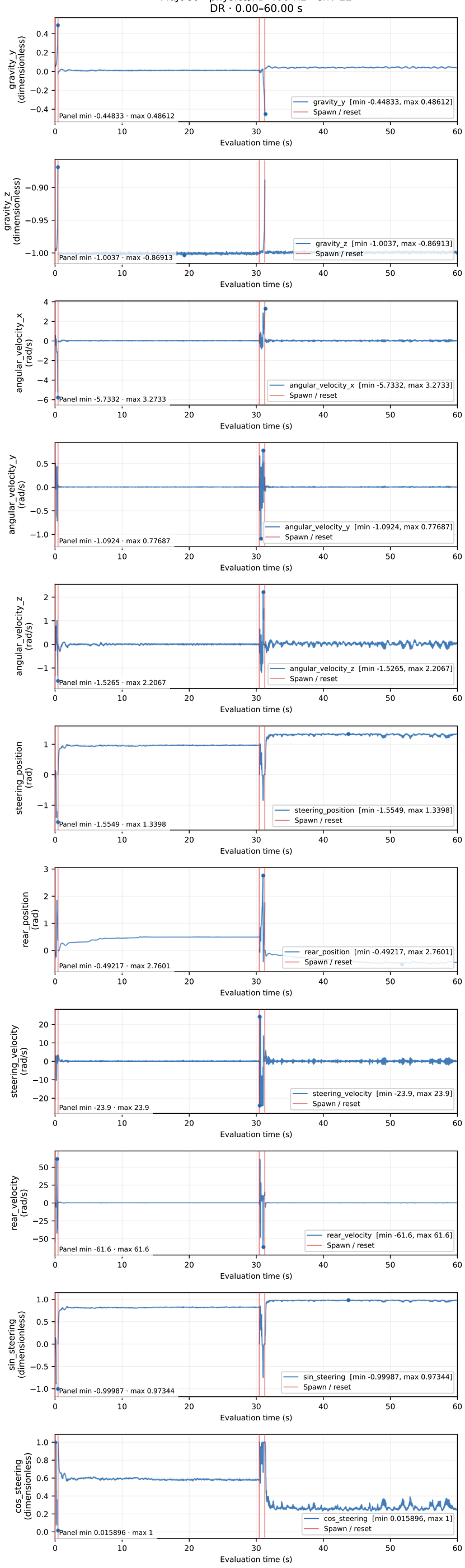
```
SAMPLED DR: env 11; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.88) s:
  mass x [0.804451, 1.08756, 0.932451, 1.17324]
  CoM XYZ mm [-3.95175, 1.66437, 3.19254]
  mu_s [0.69316, 0.749678, 0.616244, 0.669571]
  mu_d [0.69316, 0.741429, 0.586498, 0.669571]
  Kp [steer,rear] [10.5376, 0]; Kd [0.192053, 0]
  spawn roll -0.0839723 rad; rate 0.152834 rad/s
  gyro bias XYZ rad/s [1.32216e-05, -0.000139725, -3.55158e-05]
  gravity bias YZ [0.000531016, -0.00231086]
E2 [0.88, 30.88) s:
  mass x [0.976272, 0.885468, 0.968607, 1.10736]
  CoM XYZ mm [0.554094, 4.28003, 4.62543]
  mu_s [0.749678, 0.568073, 0.700521, 0.538976]
  mu_d [0.741429, 0.568073, 0.641329, 0.538976]
  Kp [steer,rear] [10.8778, 0]; Kd [0.195074, 0]
  spawn roll 0.00723343 rad; rate 0.204298 rad/s
  gyro bias XYZ rad/s [0.000249439, 4.50987e-05, -7.61758e-05]
  gravity bias YZ [0.00256586, -0.00277735]
E3 [30.88, 60.00) s:
  mass x [1.06718, 1.15346, 0.838004, 0.862932]
  CoM XYZ mm [-2.41201, -3.27271, 4.92665]
  mu_s [0.568073, 0.757579, 0.71155, 0.538434]
  mu_d [0.568073, 0.70921, 0.71155, 0.538434]
  Kp [steer,rear] [9.21777, 0]; Kd [0.192868, 0]
  spawn roll -0.0825202 rad; rate 0.213405 rad/s
  gyro bias XYZ rad/s [4.13111e-05, 0.000199249, -0.00012153]
  gravity bias YZ [0.00149148, 0.00141227]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 12 · full 60 s

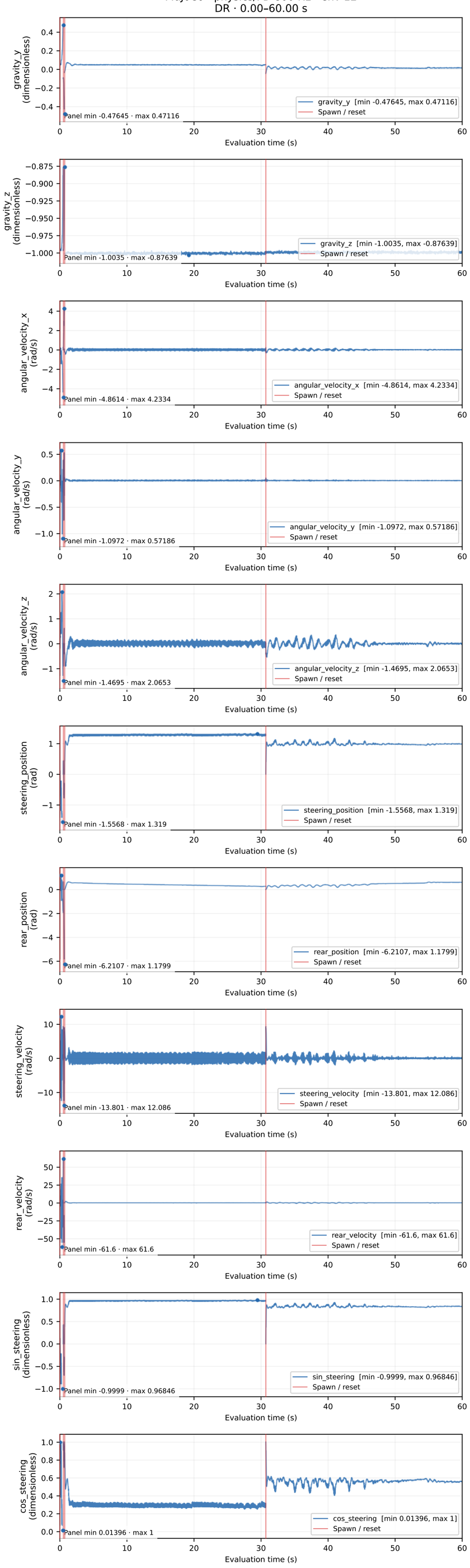
Isaac Lab · physics/PD 400 Hz · env 12
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 12
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 12
DR · 0.00–60.00 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

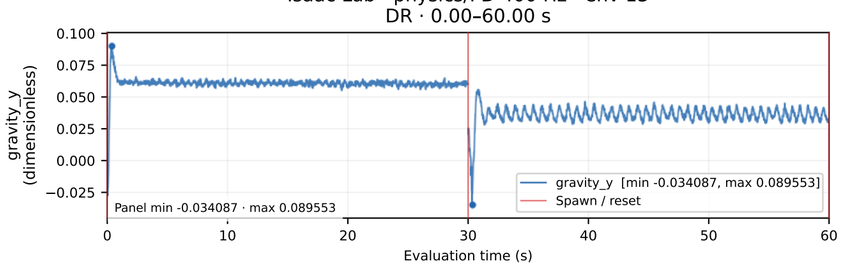
```
SAMPLED DR: env 12; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.843677, 1.04398, 0.870024, 1.13626]
  CoM XYZ mm [2.36219, 0.202864, 1.9197]
  mu_s [0.693205, 0.726175, 0.632409, 0.576972]
  mu_d [0.693205, 0.558574, 0.589076, 0.576972]
  Kp [steer,rear] [10.8603, 0]; Kd [0.20193, 0]
  spawn roll -0.00308955 rad; rate -0.0558987 rad/s
  gyro bias XYZ rad/s [1.09179e-05, -0.00014741, 7.55158e-05]
  gravity bias YZ [-0.00297523, 0.00145696]
E2 [30.00, 60.00) s:
  mass x [0.844162, 0.909308, 0.829776, 1.01903]
  CoM XYZ mm [-2.25903, -1.72129, 2.47356]
  mu_s [0.787792, 0.533455, 0.703711, 0.659445]
  mu_d [0.617134, 0.533455, 0.703711, 0.547619]
  Kp [steer,rear] [10.3056, 0]; Kd [0.210959, 0]
  spawn roll 0.00288594 rad; rate 0.0706386 rad/s
  gyro bias XYZ rad/s [0.000319707, 0.000310775, -0.000162301]
  gravity bias YZ [0.000377707, -0.00287623]
Six significant digits shown; full precision: sampled_dr_values.json.
```

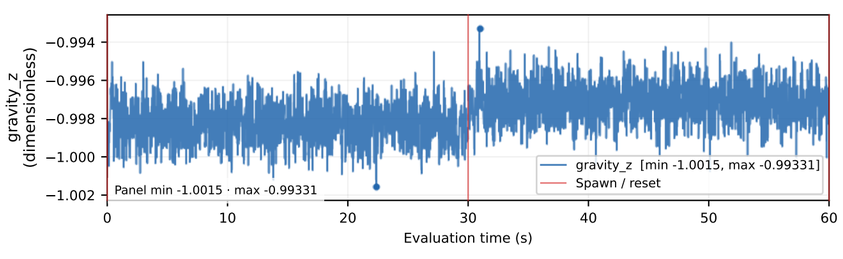
```
SAMPLED DR: env 12; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.44) s:
  mass x [0.888981, 1.00743, 0.831862, 1.03098]
  CoM XYZ mm [-4.1388, -0.447105, 1.67687]
  mu_s [0.733515, 0.538434, 0.704749, 0.541939]
  mu_d [0.558392, 0.538434, 0.541926, 0.534359]
  Kp [steer,rear] [10.8632, 0]; Kd [0.208468, 0]
  spawn roll -0.057985 rad; rate 0.081589 rad/s
  gyro bias XYZ rad/s [-0.00016212, -4.68457e-05, 1.47177e-05]
  gravity bias YZ [-0.00321399, 0.00226372]
E2 [0.44, 30.44) s:
  mass x [1.0346, 1.12616, 1.17533, 1.01379]
  CoM XYZ mm [-4.71101, 1.97284, 1.61729]
  mu_s [0.640016, 0.791209, 0.528253, 0.733515]
  mu_d [0.513141, 0.767936, 0.528253, 0.558392]
  Kp [steer,rear] [10.2496, 0]; Kd [0.217376, 0]
  spawn roll 0.0268486 rad; rate -0.112248 rad/s
  gyro bias XYZ rad/s [5.21292e-05, -0.000253077, -6.43064e-05]
  gravity bias YZ [0.00124444, -0.000723726]
E3 [30.44, 31.28) s:
  mass x [0.895664, 0.861537, 1.01434, 0.801483]
  CoM XYZ mm [-0.735719, 1.39525, -0.138491]
  mu_s [0.616244, 0.633024, 0.568073, 0.640016]
  mu_d [0.586498, 0.568172, 0.568073, 0.513141]
  Kp [steer,rear] [10.0039, 0]; Kd [0.198646, 0]
  spawn roll 0.0119817 rad; rate -0.0276258 rad/s
  gyro bias XYZ rad/s [-0.000283343, 0.000158501, -1.02695e-05]
  gravity bias YZ [0.00112562, 0.00173186]
E4 [31.28, 60.00) s:
  mass x [0.810855, 1.19466, 1.01735, 1.05576]
  CoM XYZ mm [-3.28053, -2.79567, 3.18595]
  mu_s [0.732187, 0.71155, 0.640867, 0.637675]
  mu_d [0.631664, 0.71155, 0.556841, 0.637675]
  Kp [steer,rear] [9.59382, 0]; Kd [0.180522, 0]
  spawn roll -0.0237513 rad; rate 0.0525132 rad/s
  gyro bias XYZ rad/s [-0.000118069, -1.40752e-05, -3.04469e-05]
  gravity bias YZ [-0.00029513, 0.000152421]
Six significant digits shown; full precision: sampled_dr_values.json.
```

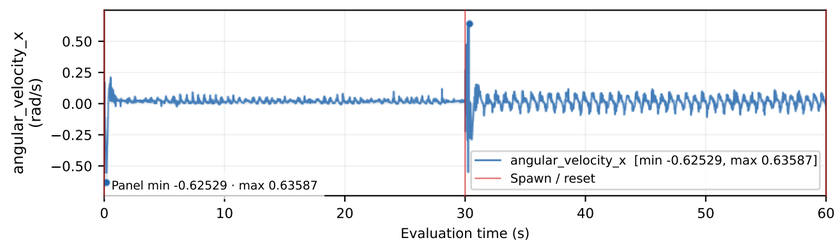
```
SAMPLED DR: env 12; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.52) s:
  mass x [0.888981, 1.00743, 0.831862, 1.03098]
  CoM XYZ mm [-4.1388, -0.447105, 1.67687]
  mu_s [0.733515, 0.538434, 0.704749, 0.541939]
  mu_d [0.558392, 0.538434, 0.541926, 0.534359]
  Kp [steer,rear] [10.8632, 0]; Kd [0.208468, 0]
  spawn roll -0.057985 rad; rate 0.081589 rad/s
  gyro bias XYZ rad/s [-0.00016212, -4.68457e-05, 1.47177e-05]
  gravity bias YZ [-0.00321399, 0.00226372]
E2 [0.52, 0.72) s:
  mass x [0.890015, 1.17945, 1.10153, 1.06853]
  CoM XYZ mm [1.44741, 2.8022, -2.6186]
  mu_s [0.669571, 0.528253, 0.538976, 0.538976]
  mu_d [0.669571, 0.528253, 0.538976, 0.538976]
  Kp [steer,rear] [9.51674, 0]; Kd [0.19014, 0]
  spawn roll 0.0995909 rad; rate 0.23022 rad/s
  gyro bias XYZ rad/s [2.73531e-05, -4.06514e-05, 0.000142005]
  gravity bias YZ [0.000861103, 0.000938522]
E3 [0.72, 30.72) s:
  mass x [0.891332, 0.956026, 1.09053, 0.994289]
  CoM XYZ mm [-1.66715, -3.01877, 2.69457]
  mu_s [0.723429, 0.559972, 0.757579, 0.538434]
  mu_d [0.723429, 0.502209, 0.70921, 0.538434]
  Kp [steer,rear] [9.18502, 0]; Kd [0.215514, 0]
  spawn roll 0.0384913 rad; rate 0.0731196 rad/s
  gyro bias XYZ rad/s [-0.000197728, -0.000124193, -0.000269497]
  gravity bias YZ [0.00145792, -0.00140849]
E4 [30.72, 60.00) s:
  mass x [1.14791, 1.01322, 1.08839, 1.01143]
  CoM XYZ mm [-2.36312, 1.30928, 0.963934]
  mu_s [0.528253, 0.723429, 0.71008, 0.733515]
  mu_d [0.528253, 0.723429, 0.59371, 0.558392]
  Kp [steer,rear] [9.61071, 0]; Kd [0.186415, 0]
  spawn roll 0.0462045 rad; rate 0.218468 rad/s
  gyro bias XYZ rad/s [-0.000102087, -0.000248652, -7.2445e-05]
  gravity bias YZ [0.000117711, 0.000915891]
Six significant digits shown; full precision: sampled_dr_values.json.
```

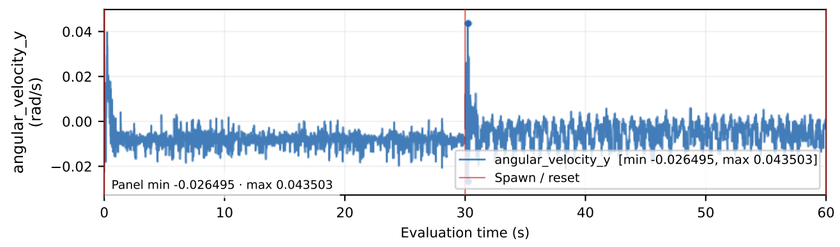
model_2500 · All 11 actual policy observations · env 13 · full 60 s

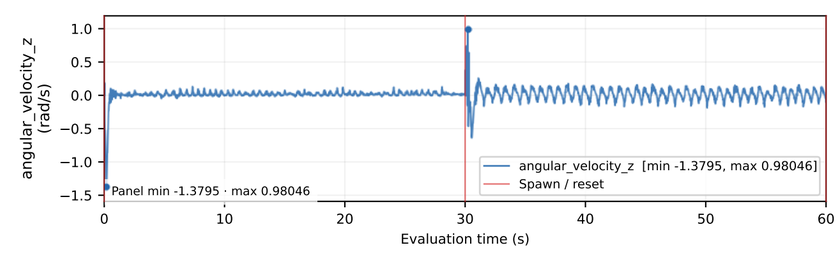
Isaac Lab · physics/PD 400 Hz · env 13
DR · 0.00–60.00 s

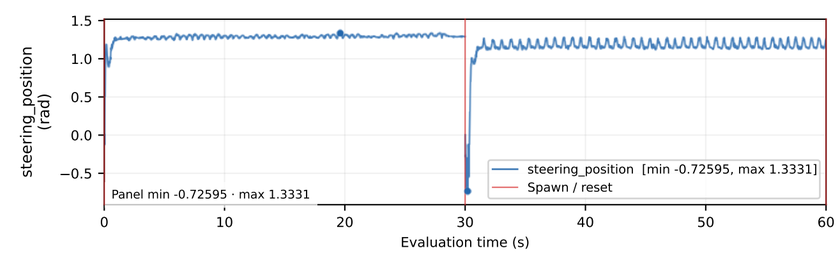


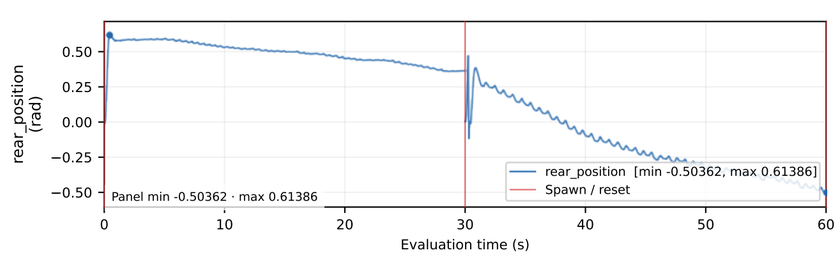


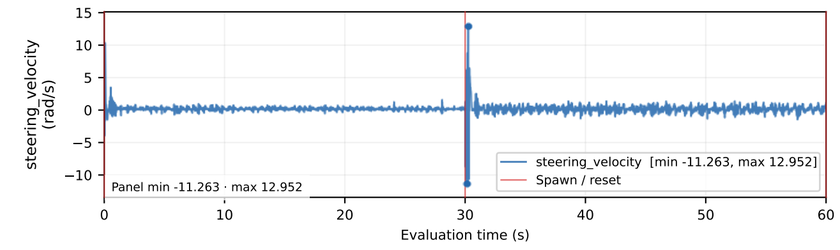


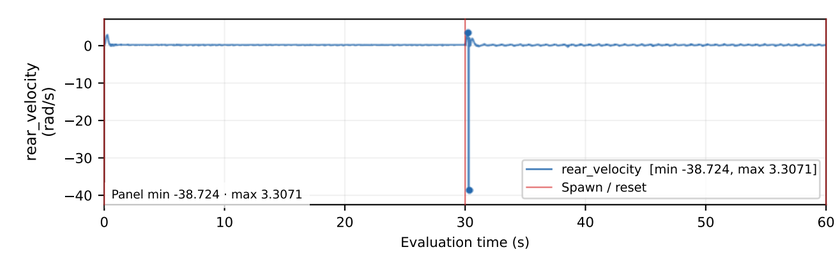


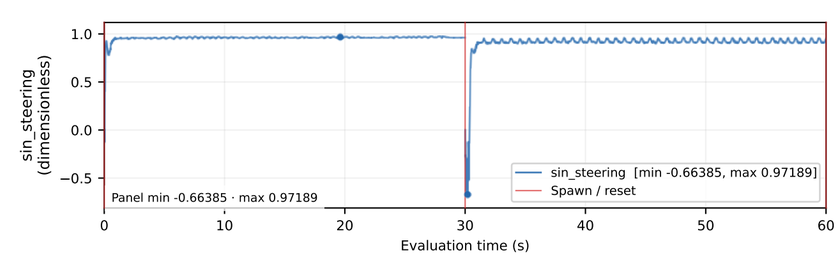


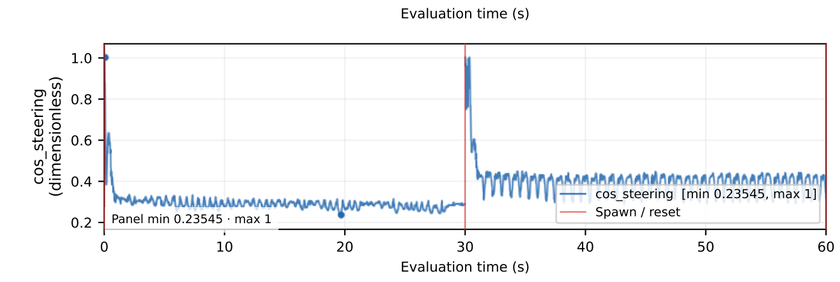




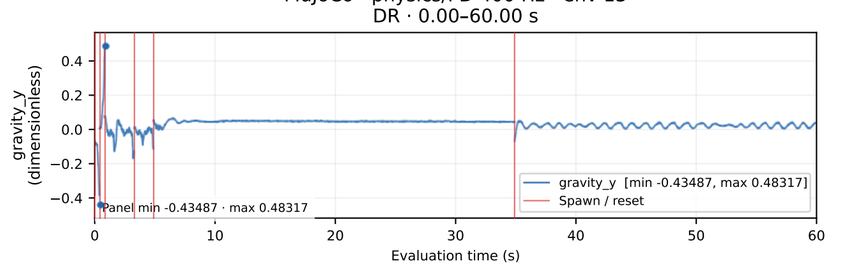


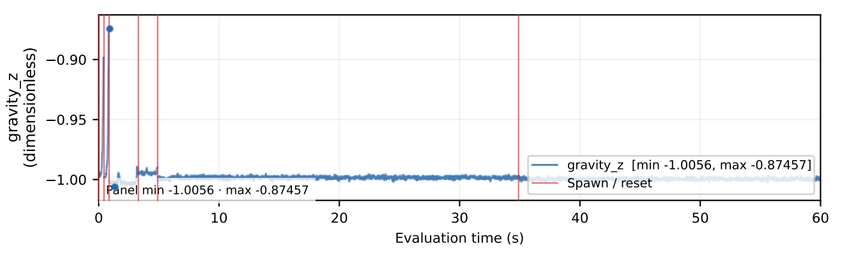


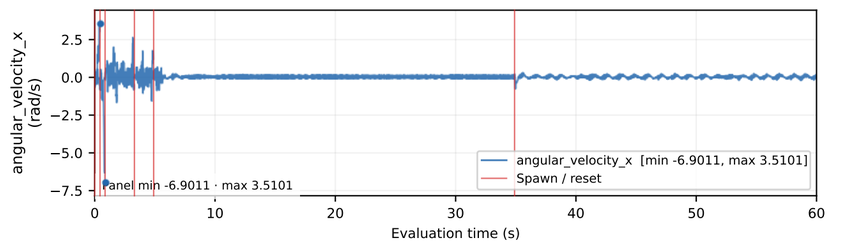


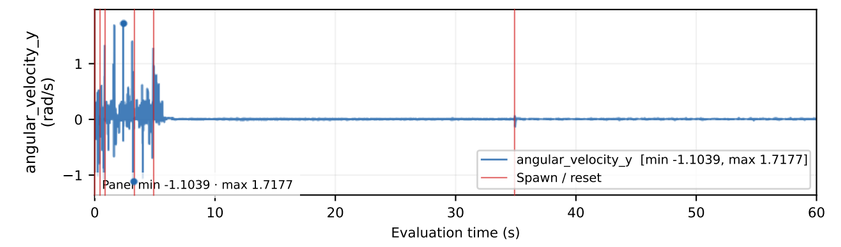


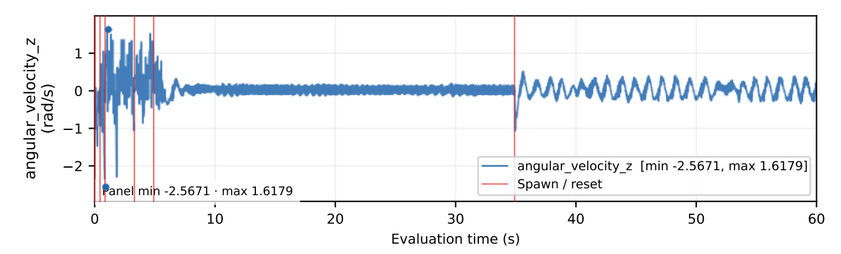
MuJoCo · physics/PD 400 Hz · env 13
DR · 0.00–60.00 s

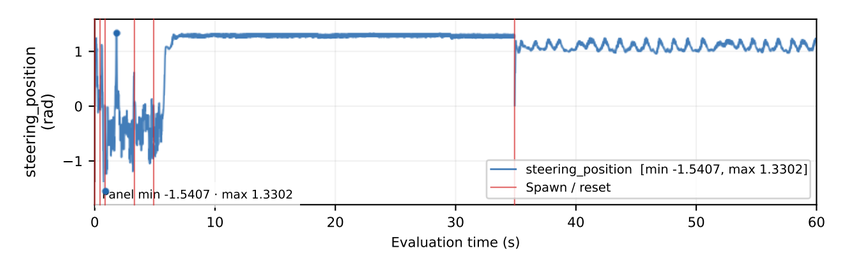


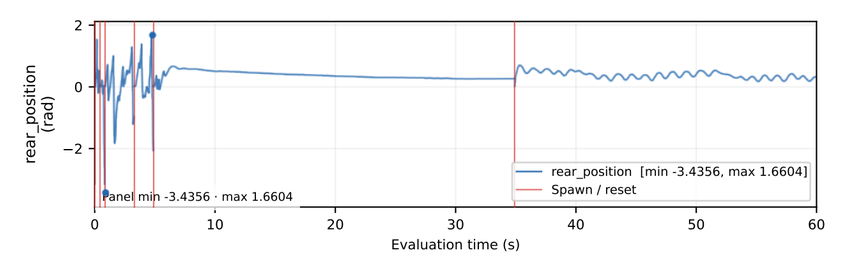


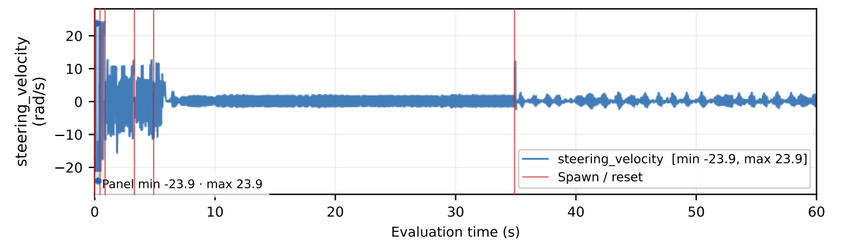


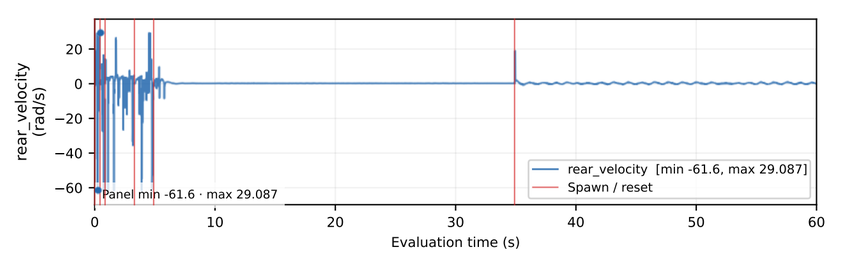


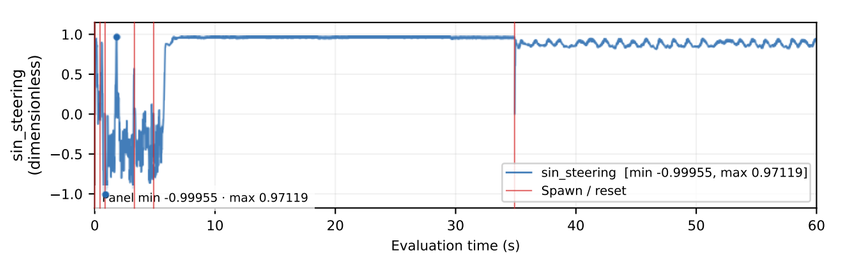


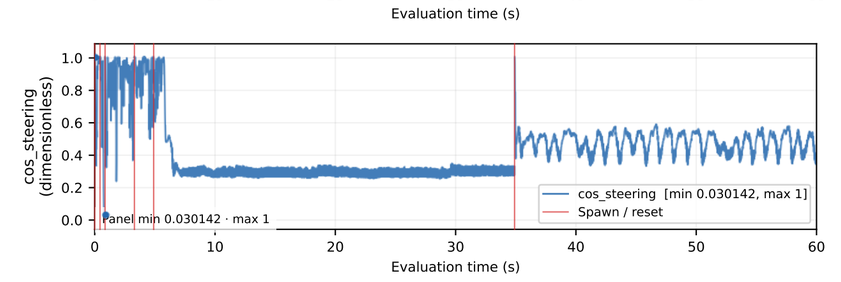




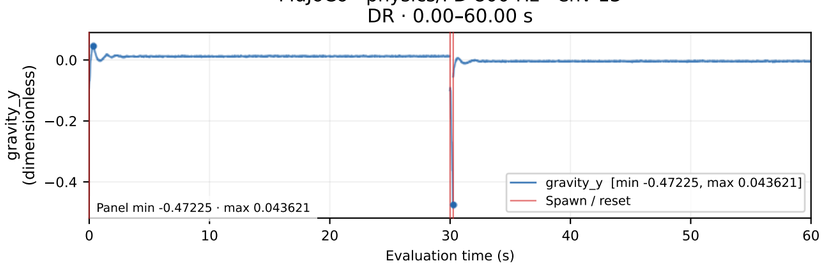


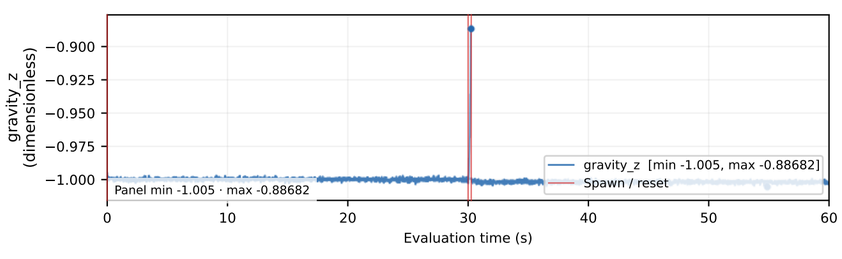


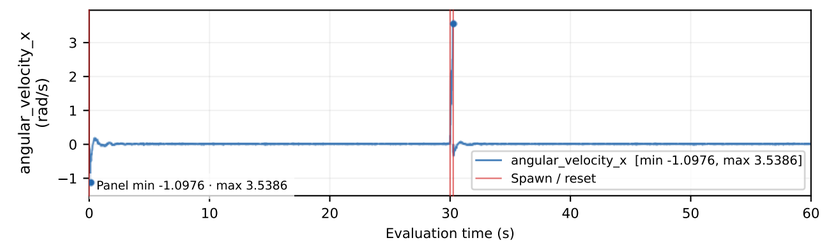


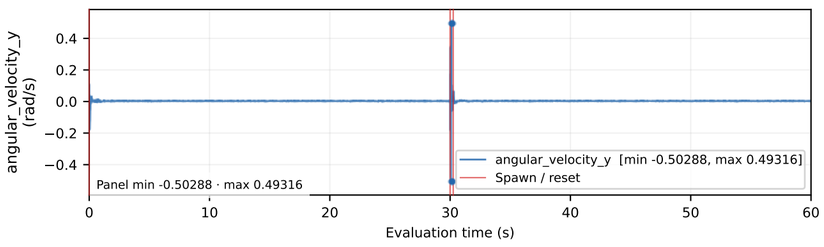


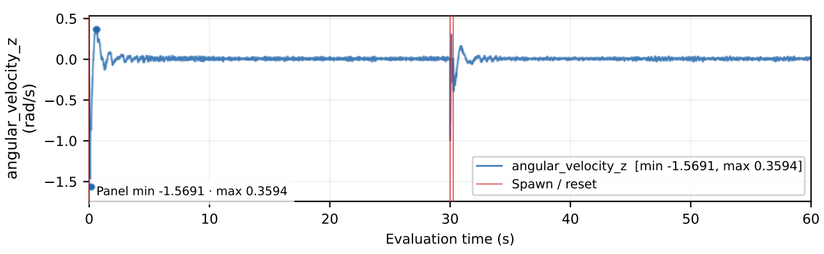
MuJoCo · physics/PD 800 Hz · env 13
DR · 0.00–60.00 s

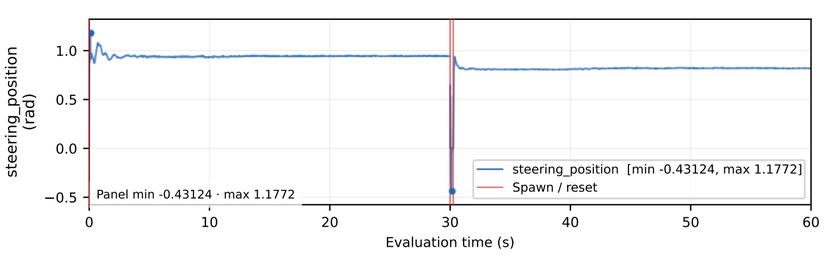


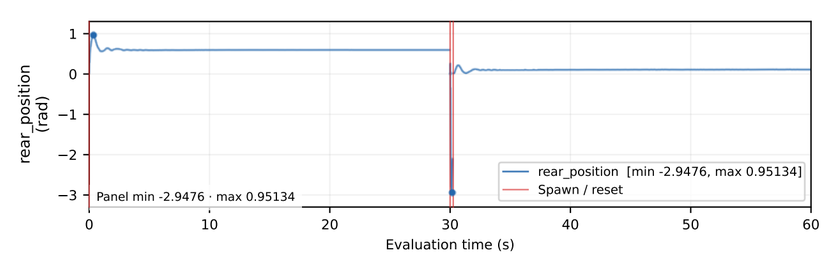


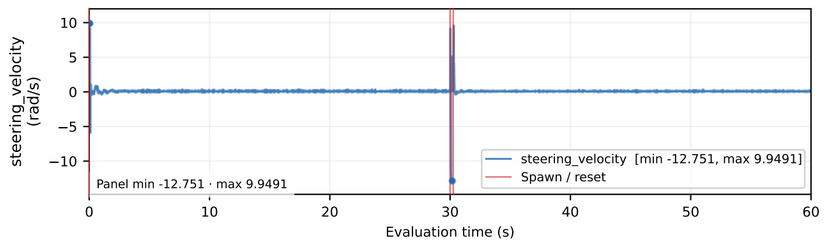


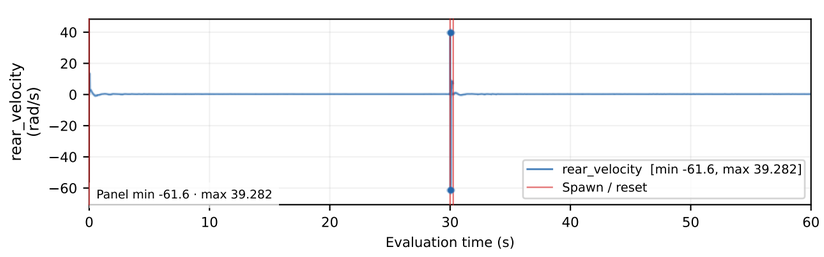


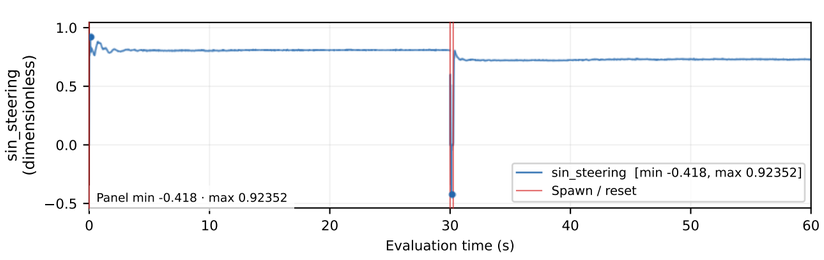


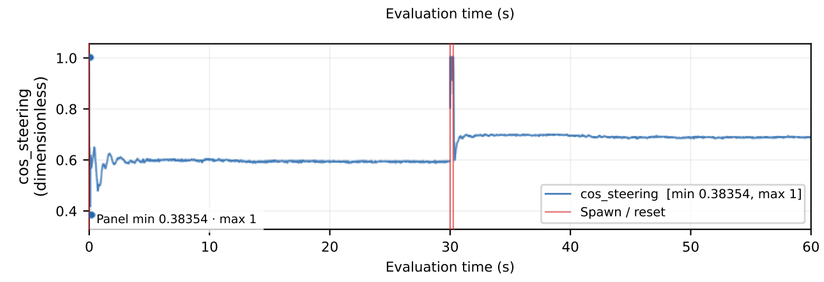












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

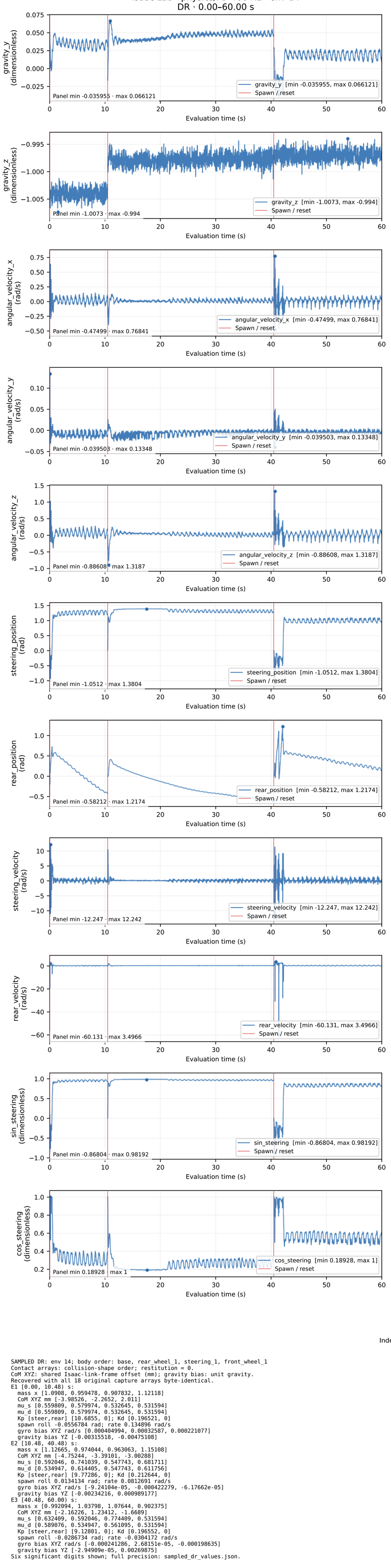
```
SAMPLED DR: env 13; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.14819, 1.09148, 0.901625, 0.866687]
  CoM XYZ mm [-1.49745, -4.85852, 4.10487]
  mu_s [0.576972, 0.764681, 0.703711, 0.760821]
  mu_d [0.576972, 0.764681, 0.703711, 0.670315]
  Kp [steer,rear] [9.5585, 0]; Kd [0.19592, 0]
  spawn roll 0.0150582 rad; rate -0.209008 rad/s
  gyro bias XYZ rad/s [-0.0002458, 0.000257238, 9.44563e-05]
  gravity bias YZ [-0.00107082, -0.000198226]
E2 [30.00, 60.00) s:
  mass x [1.12365, 1.02316, 0.969298, 0.904739]
  CoM XYZ mm [4.35705, -0.931233, 3.05012]
  mu_s [0.598343, 0.628821, 0.713129, 0.659445]
  mu_d [0.598343, 0.628821, 0.713129, 0.547619]
  Kp [steer,rear] [10.1481, 0]; Kd [0.187719, 0]
  spawn roll -0.0219507 rad; rate -0.224828 rad/s
  gyro bias XYZ rad/s [-0.000266752, -4.27344e-05, 0.000211438]
  gravity bias YZ [-0.000458392, 0.00216717]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 13; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.44) s:
  mass x [1.1329, 1.14826, 0.887345, 0.892039]
  CoM XYZ mm [-0.072569, 2.53223, 2.04232]
  mu_s [0.728342, 0.700521, 0.559972, 0.528253]
  mu_d [0.728342, 0.641329, 0.502209, 0.528253]
  Kp [steer,rear] [9.98219, 0]; Kd [0.204077, 0]
  spawn roll 0.0961499 rad; rate -0.0648344 rad/s
  gyro bias XYZ rad/s [-8.41829e-05, 9.3231e-05, -0.000287958]
  gravity bias YZ [0.00221957, 8.91091e-05]
E2 [0.44, 0.86) s:
  mass x [1.18048, 1.10671, 0.913639, 0.89274]
  CoM XYZ mm [-0.235183, 4.12978, -3.9335]
  mu_s [0.611239, 0.757579, 0.616244, 0.597748]
  mu_d [0.611239, 0.70921, 0.586498, 0.597748]
  Kp [steer,rear] [9.10023, 0]; Kd [0.215477, 0]
  spawn roll -0.00256985 rad; rate -0.131061 rad/s
  gyro bias XYZ rad/s [7.04588e-05, -3.01761e-05, -0.000176978]
  gravity bias YZ [0.000567873, 0.00103235]
E3 [0.86, 3.30) s:
  mass x [1.16124, 0.824091, 1.11037, 1.07]
  CoM XYZ mm [2.12359, 2.17524, -0.110732]
  mu_s [0.666375, 0.640016, 0.704749, 0.748289]
  mu_d [0.519145, 0.513141, 0.541926, 0.689499]
  Kp [steer,rear] [9.85806, 0]; Kd [0.186709, 0]
  spawn roll -0.0667805 rad; rate -0.0786676 rad/s
  gyro bias XYZ rad/s [0.00046598, 9.39327e-05, 0.000110484]
  gravity bias YZ [0.00476382, -0.00345774]
E4 [3.30, 4.90) s:
  mass x [0.889313, 0.813812, 1.09052, 1.05501]
  CoM XYZ mm [1.75968, 3.00834, -2.88236]
  mu_s [0.640867, 0.597748, 0.559972, 0.71155]
  mu_d [0.556841, 0.597748, 0.502209, 0.71155]
  Kp [steer,rear] [9.80407, 0]; Kd [0.190426, 0]
  spawn roll -0.00749381 rad; rate -0.0256039 rad/s
  gyro bias XYZ rad/s [-4.49526e-05, 0.000359247, 0.000179771]
  gravity bias YZ [-0.00136921, 0.00503738]
E5 [4.90, 34.90) s:
  mass x [1.16635, 0.964278, 0.891603, 1.12673]
  CoM XYZ mm [1.93179, -2.62501, -4.16081]
  mu_s [0.723429, 0.71008, 0.727426, 0.616244]
  mu_d [0.723429, 0.59371, 0.606358, 0.586498]
  Kp [steer,rear] [9.41256, 0]; Kd [0.201394, 0]
  spawn roll -0.0488085 rad; rate -0.0852721 rad/s
  gyro bias XYZ rad/s [-0.000502116, -3.32743e-05, 0.000103514]
  gravity bias YZ [0.000722132, 0.000688708]
E6 [34.90, 60.00) s:
  mass x [0.90303, 1.0851, 0.867758, 1.0553]
  CoM XYZ mm [-2.16255, 0.399354, 1.11696]
  mu_s [0.611239, 0.732187, 0.749678, 0.733515]
  mu_d [0.611239, 0.631664, 0.741429, 0.558392]
  Kp [steer,rear] [9.24214, 0]; Kd [0.181495, 0]
  spawn roll 0.0723589 rad; rate 0.164877 rad/s
  gyro bias XYZ rad/s [-0.000108821, -0.000119695, -2.92104e-05]
  gravity bias YZ [0.000176065, 9.40162e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

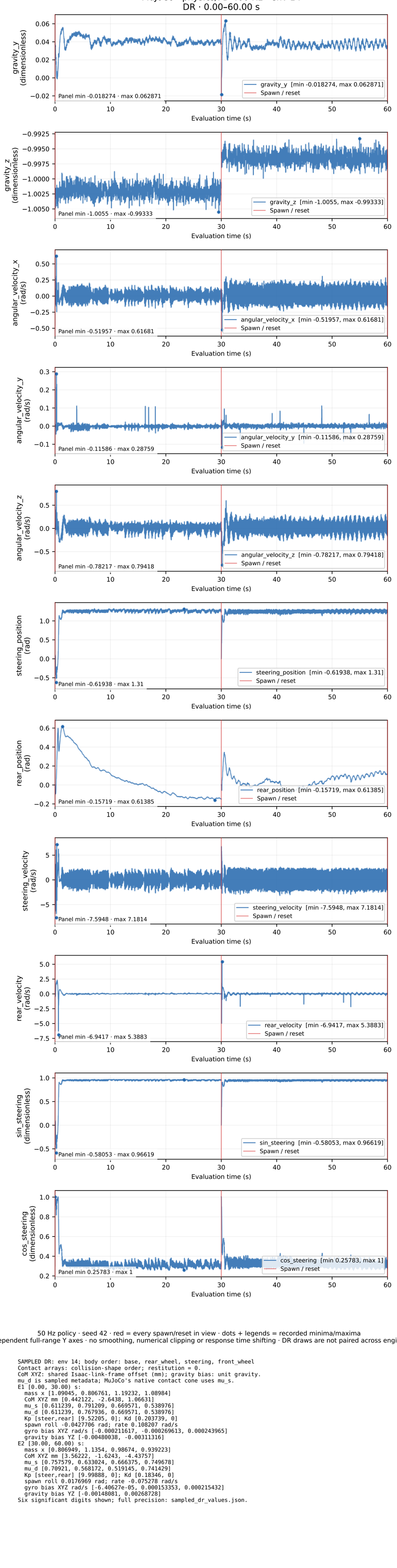
```
SAMPLED DR: env 13; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.1329, 1.14826, 0.887345, 0.892039]
  CoM XYZ mm [-0.072569, 2.53223, 2.04232]
  mu_s [0.728342, 0.700521, 0.559972, 0.528253]
  mu_d [0.728342, 0.641329, 0.502209, 0.528253]
  Kp [steer,rear] [9.98219, 0]; Kd [0.204077, 0]
  spawn roll 0.0961499 rad; rate -0.0648344 rad/s
  gyro bias XYZ rad/s [-8.41829e-05, 9.3231e-05, -0.000287958]
  gravity bias YZ [0.00221957, 8.91091e-05]
E2 [30.00, 30.26) s:
  mass x [1.07626, 0.853924, 0.874087, 0.866441]
  CoM XYZ mm [-1.51051, -2.57644, -2.51735]
  mu_s [0.733515, 0.791209, 0.597748, 0.728342]
  mu_d [0.558392, 0.767936, 0.597748, 0.728342]
  Kp [steer,rear] [10.4238, 0]; Kd [0.186222, 0]
  spawn roll 0.0973156 rad; rate 0.215546 rad/s
  gyro bias XYZ rad/s [-8.64701e-05, -2.65843e-05, 5.33129e-05]
  gravity bias YZ [-0.00274842, -0.00384582]
E3 [30.26, 60.00) s:
  mass x [0.956472, 0.884212, 1.03474, 0.989246]
  CoM XYZ mm [-0.261673, 4.47667, -0.31183]
  mu_s [0.732187, 0.791209, 0.749678, 0.757579]
  mu_d [0.631664, 0.767936, 0.741429, 0.709211]
  Kp [steer,rear] [9.13301, 0]; Kd [0.190991, 0]
  spawn roll 0.0554458 rad; rate -0.237323 rad/s
  gyro bias XYZ rad/s [-0.000265184, 0.000105029, -0.000140464]
  gravity bias YZ [-0.000820023, -0.001881164]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 14 · full 60 s

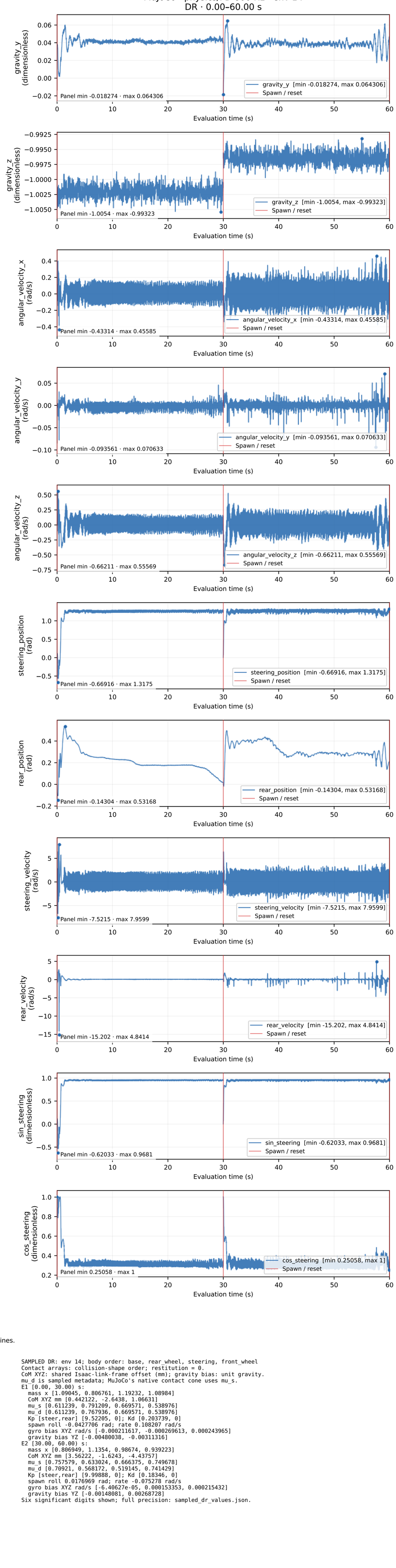
Isaac Lab · physics/PD 400 Hz · env 14
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 14
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 14
DR · 0.00–60.00 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

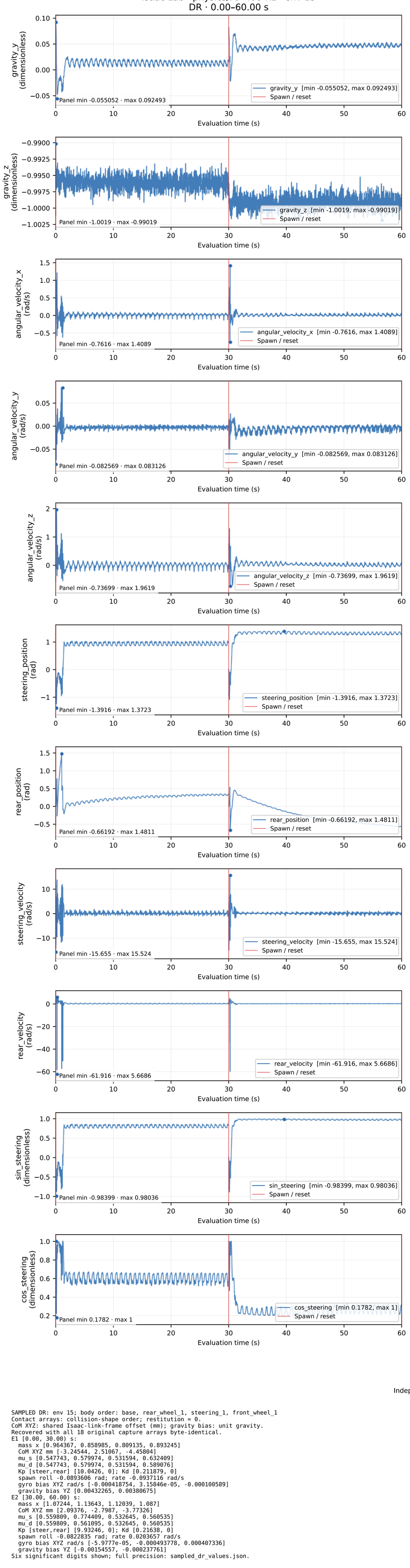
```
SAMPLED DR: env 14; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 10.48) s:
  mass x [1.0908, 0.959478, 0.907832, 1.12118]
  CoM XYZ mm [-3.98526, -2.2652, 2.011]
  mu_s [0.559809, 0.579974, 0.532645, 0.531594]
  mu_d [0.559809, 0.579974, 0.532645, 0.531594]
  Kp [steer,rear] [10.6855, 0]; Kd [0.196521, 0]
  spawn roll -0.0556784 rad; rate 0.134896 rad/s
  gyro bias XYZ rad/s [0.000404994, 0.00032587, 0.000221077]
  gravity bias YZ [-0.00315518, -0.00475108]
E2 [10.48, 40.48) s:
  mass x [1.12665, 0.974044, 0.963063, 1.15108]
  CoM XYZ mm [-4.75244, -3.39101, -3.00288]
  mu_s [0.592046, 0.741039, 0.547743, 0.681711]
  mu_d [0.534947, 0.614405, 0.547743, 0.611756]
  Kp [steer,rear] [9.77286, 0]; Kd [0.212644, 0]
  spawn roll 0.0134134 rad; rate 0.0812691 rad/s
  gyro bias XYZ rad/s [-9.24104e-05, -0.000422279, -6.17662e-05]
  gravity bias YZ [-0.00234216, 0.000989177]
E3 [40.48, 60.00) s:
  mass x [0.992094, 1.03798, 1.07644, 0.902375]
  CoM XYZ mm [-2.16226, 1.23412, -1.6689]
  mu_s [0.632409, 0.592046, 0.774409, 0.531594]
  mu_d [0.589076, 0.534947, 0.561095, 0.531594]
  Kp [steer,rear] [9.12801, 0]; Kd [0.196552, 0]
  spawn roll -0.0286734 rad; rate -0.0304172 rad/s
  gyro bias XYZ rad/s [-0.000241286, 2.68151e-05, -0.000198635]
  gravity bias YZ [-2.94909e-05, 0.00269875]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 14; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09045, 0.806761, 1.19232, 1.08984]
  CoM XYZ mm [0.442122, -2.6438, 1.06631]
  mu_s [0.611239, 0.791209, 0.669571, 0.538976]
  mu_d [0.611239, 0.767936, 0.669571, 0.538976]
  Kp [steer,rear] [9.52205, 0]; Kd [0.203739, 0]
  spawn roll -0.0427706 rad; rate 0.108207 rad/s
  gyro bias XYZ rad/s [-0.000211617, -0.000269613, 0.000243965]
  gravity bias YZ [-0.00480038, -0.00311316]
E2 [30.00, 60.00) s:
  mass x [0.806949, 1.1354, 0.98674, 0.939223]
  CoM XYZ mm [3.56222, -1.6243, -4.43757]
  mu_s [0.757579, 0.633024, 0.666375, 0.749678]
  mu_d [0.70921, 0.568172, 0.519145, 0.741429]
  Kp [steer,rear] [9.99888, 0]; Kd [0.18346, 0]
  spawn roll 0.0176969 rad; rate -0.075278 rad/s
  gyro bias XYZ rad/s [-6.40627e-05, 0.000153353, 0.000215432]
  gravity bias YZ [-0.00148081, 0.00268728]
Six significant digits shown; full precision: sampled_dr_values.json.
```

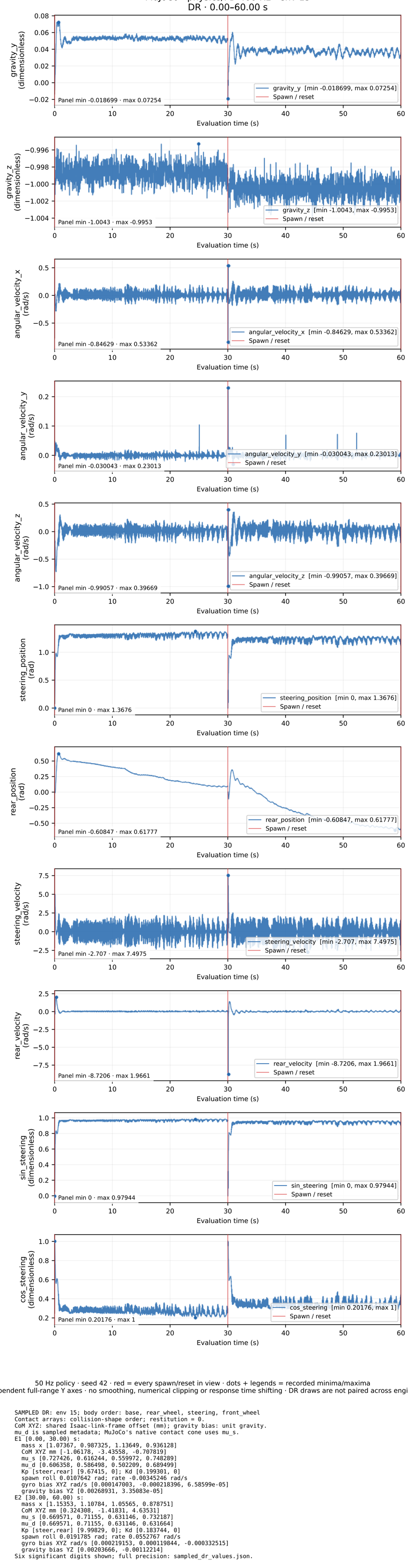
```
SAMPLED DR: env 14; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09045, 0.806761, 1.19232, 1.08984]
  CoM XYZ mm [0.442122, -2.6438, 1.06631]
  mu_s [0.611239, 0.791209, 0.669571, 0.538976]
  mu_d [0.611239, 0.767936, 0.669571, 0.538976]
  Kp [steer,rear] [9.52205, 0]; Kd [0.203739, 0]
  spawn roll -0.0427706 rad; rate 0.108207 rad/s
  gyro bias XYZ rad/s [-0.000211617, -0.000269613, 0.000243965]
  gravity bias YZ [-0.00480038, -0.00311316]
E2 [30.00, 60.00) s:
  mass x [0.806949, 1.1354, 0.98674, 0.939223]
  CoM XYZ mm [3.56222, -1.6243, -4.43757]
  mu_s [0.757579, 0.633024, 0.666375, 0.749678]
  mu_d [0.70921, 0.568172, 0.519145, 0.741429]
  Kp [steer,rear] [9.99888, 0]; Kd [0.18346, 0]
  spawn roll 0.0176969 rad; rate -0.075278 rad/s
  gyro bias XYZ rad/s [-6.40627e-05, 0.000153353, 0.000215432]
  gravity bias YZ [-0.00148081, 0.00268728]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · All 11 actual policy observations · env 15 · full 60 s

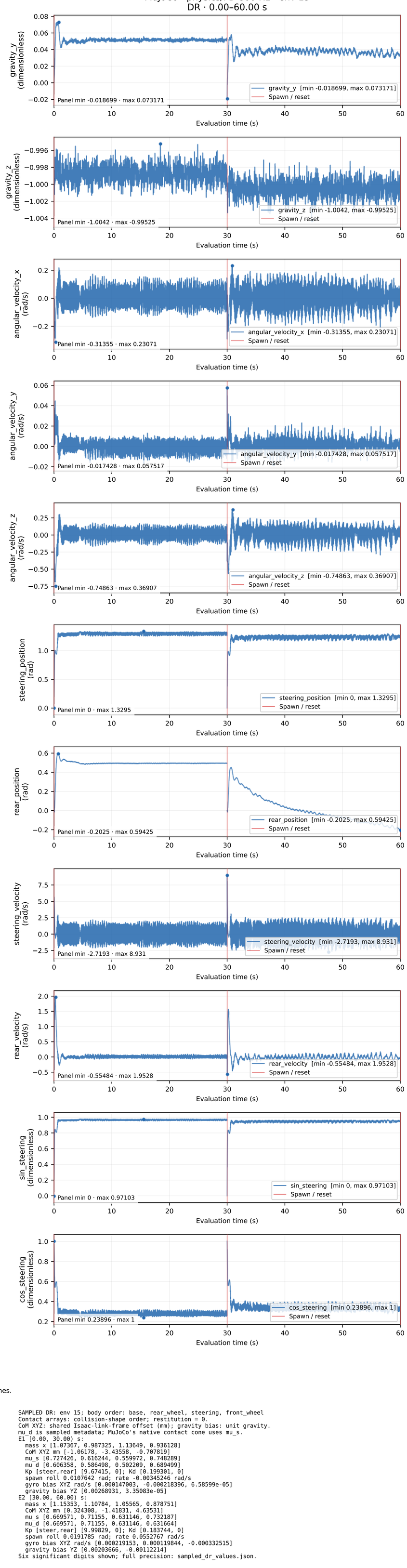
Isaac Lab · physics/PD 400 Hz · env 15
DR · 0.00–60.00 s



MuJoCo · physics/PD 400 Hz · env 15
DR · 0.00–60.00 s



MuJoCo · physics/PD 800 Hz · env 15
DR · 0.00–60.00 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 15; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.964367, 0.858985, 0.809135, 0.893245]
  CoM XYZ mm [-3.24544, 2.51067, -4.45804]
  mu_s [0.547743, 0.579974, 0.531594, 0.632409]
  mu_d [0.547743, 0.579974, 0.531594, 0.589076]
  Kp [steer,rear] [10.0426, 0]; Kd [0.211879, 0]
  spawn roll -0.0893606 rad; rate -0.0937116 rad/s
  gyro bias XYZ rad/s [-0.000418754, 3.15846e-05, -0.000100589]
  gravity bias YZ [0.00432265, 0.00380675]
E2 [30.00, 60.00) s:
  mass x [1.07244, 1.13643, 1.12039, 1.087]
  CoM XYZ mm [2.09376, -2.7987, -3.77326]
  mu_s [0.559809, 0.774409, 0.532645, 0.560535]
  mu_d [0.559809, 0.561095, 0.532645, 0.560535]
  Kp [steer,rear] [9.93246, 0]; Kd [0.21638, 0]
  spawn roll -0.0822835 rad; rate 0.0203657 rad/s
  gyro bias XYZ rad/s [-5.9777e-05, -0.000493778, 0.000407336]
  gravity bias YZ [-0.00154557, -0.000237761]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 15; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.07367, 0.987325, 1.13649, 0.936128]
  CoM XYZ mm [-1.06178, -3.43558, -0.707819]
  mu_s [0.727426, 0.616244, 0.559972, 0.748289]
  mu_d [0.606358, 0.586498, 0.502209, 0.689499]
  Kp [steer,rear] [9.67415, 0]; Kd [0.199301, 0]
  spawn roll 0.0107642 rad; rate -0.00345246 rad/s
  gyro bias XYZ rad/s [0.000147003, -0.000218396, 6.58599e-05]
  gravity bias YZ [0.00268931, 3.35083e-05]
E2 [30.00, 60.00) s:
  mass x [1.15353, 1.10784, 1.05565, 0.878751]
  CoM XYZ mm [0.324308, -1.41831, 4.63531]
  mu_s [0.669571, 0.71155, 0.631146, 0.732187]
  mu_d [0.669571, 0.71155, 0.631146, 0.631664]
  Kp [steer,rear] [9.99829, 0]; Kd [0.183744, 0]
  spawn roll 0.0191785 rad; rate 0.0552767 rad/s
  gyro bias XYZ rad/s [0.000219153, 0.000119844, -0.000332515]
  gravity bias YZ [0.00203666, -0.00112214]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 15; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.07367, 0.987325, 1.13649, 0.936128]
  CoM XYZ mm [-1.06178, -3.43558, -0.707819]
  mu_s [0.727426, 0.616244, 0.559972, 0.748289]
  mu_d [0.606358, 0.586498, 0.502209, 0.689499]
  Kp [steer,rear] [9.67415, 0]; Kd [0.199301, 0]
  spawn roll 0.0107642 rad; rate -0.00345246 rad/s
  gyro bias XYZ rad/s [0.000147003, -0.000218396, 6.58599e-05]
  gravity bias YZ [0.00268931, 3.35083e-05]
E2 [30.00, 60.00) s:
  mass x [1.15353, 1.10784, 1.05565, 0.878751]
  CoM XYZ mm [0.324308, -1.41831, 4.63531]
  mu_s [0.669571, 0.71155, 0.631146, 0.732187]
  mu_d [0.669571, 0.71155, 0.631146, 0.631664]
  Kp [steer,rear] [9.99829, 0]; Kd [0.183744, 0]
  spawn roll 0.0191785 rad; rate 0.0552767 rad/s
  gyro bias XYZ rad/s [0.000219153, 0.000119844, -0.000332515]
  gravity bias YZ [0.00203666, -0.00112214]
Six significant digits shown; full precision: sampled_dr_values.json.
```